



Rheinmetall

THE INTEGRATED TECHNOLOGY GROUP

August 2025

Rheinmetall international

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05 AFRICA

15 AMERICAS

18 ASIA

08 AUSTRALIA

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RHEINMETALL – TECHNOLOGY FIELDS

AIR MANAGEMENT



THERMAL MANAGEMENT



ELECTRIFICATION



HYDROGEN



MOTION & CONTROL



LIGHTWEIGHT & MATERIAL



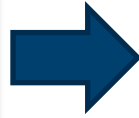
CONNECTIVITY & ADAS



Our Purpose & Vision:

- RHEINMETALL INDIA PURPOSE -

We deliver **Technology Solutions** and generate sustainable value for all **Stakeholders**.



Production PURPOSE

“ A **passionate** team delivers sustainable high performance operational metrics”



- RHEINMETALL INDIA VISION -

1. To be a market leader in offering diversified technology solutions and realize sustainable growth
2. To be a benchmark in driving excellence, promoting shared values for future

Shop Floor Management Driving Excellence on the Ground

Date: 11th August 2025

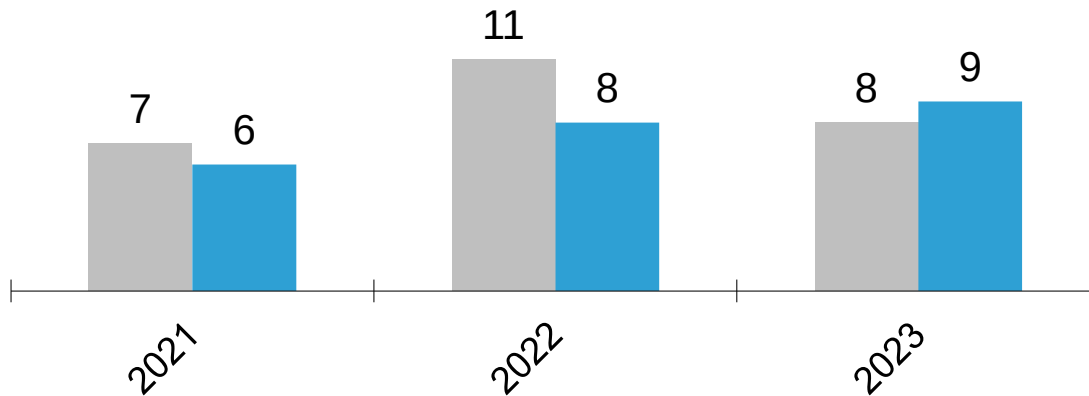
Presented by: Operation Department

Team: Santosh Wankhede, Ajay Alam, Ganesh Hole

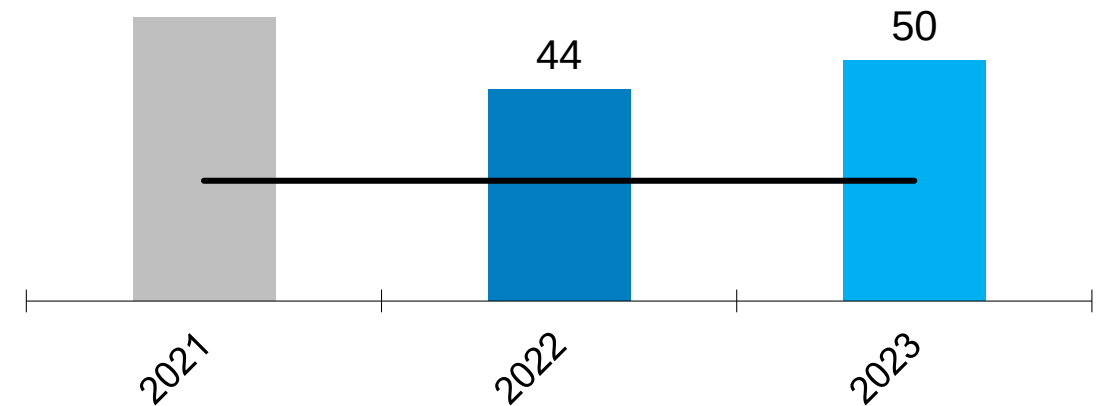
Triggers for Shop Floor Management :

- Internal: Rising defects, inconsistent quality, high downtime
- External: Customer complaints, compliance issues, market competition

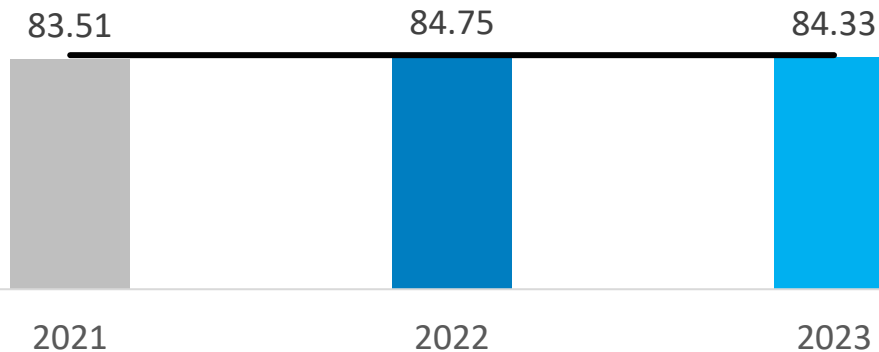
First Aid & Near Miss Incidents



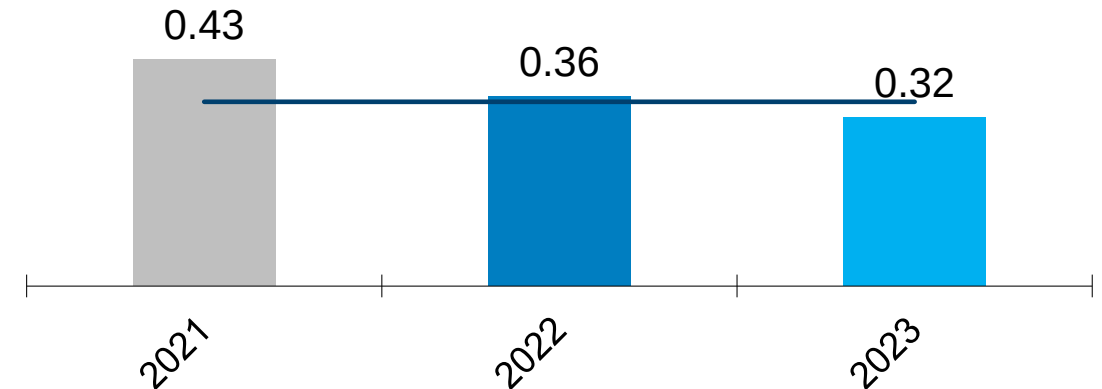
Customer Incidents Target : 25



OEE Target : 85.0 %

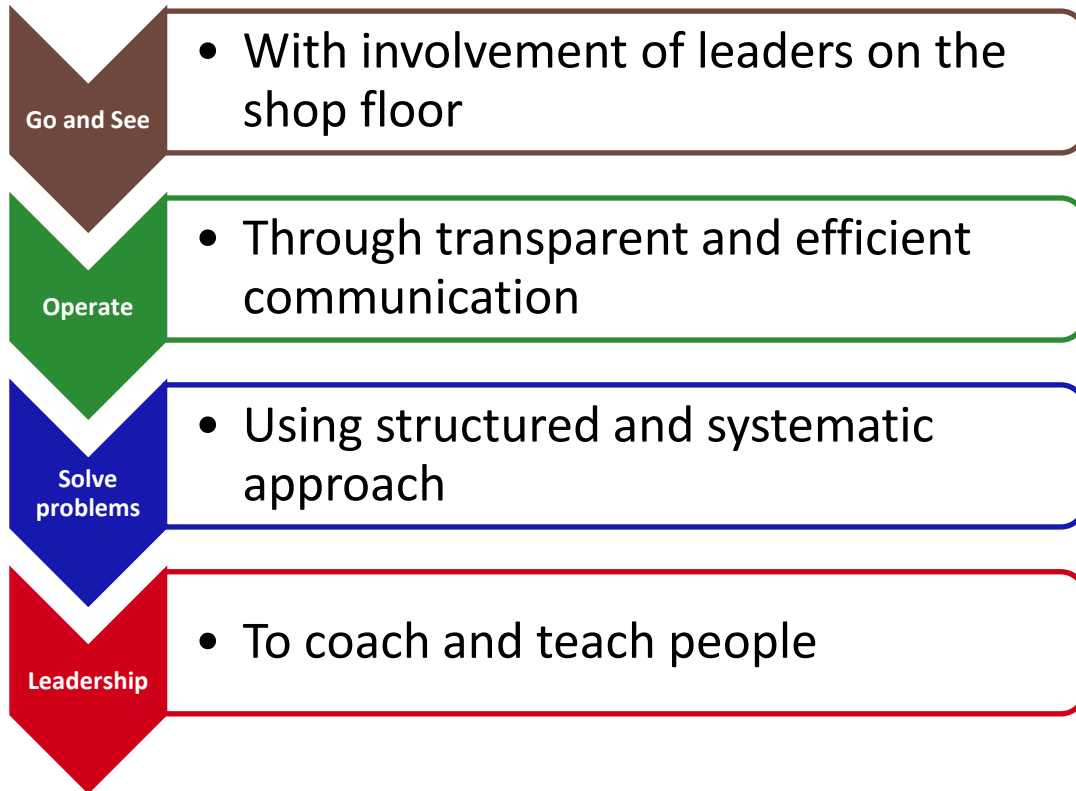


Scrap Cost Target : 0.40 %



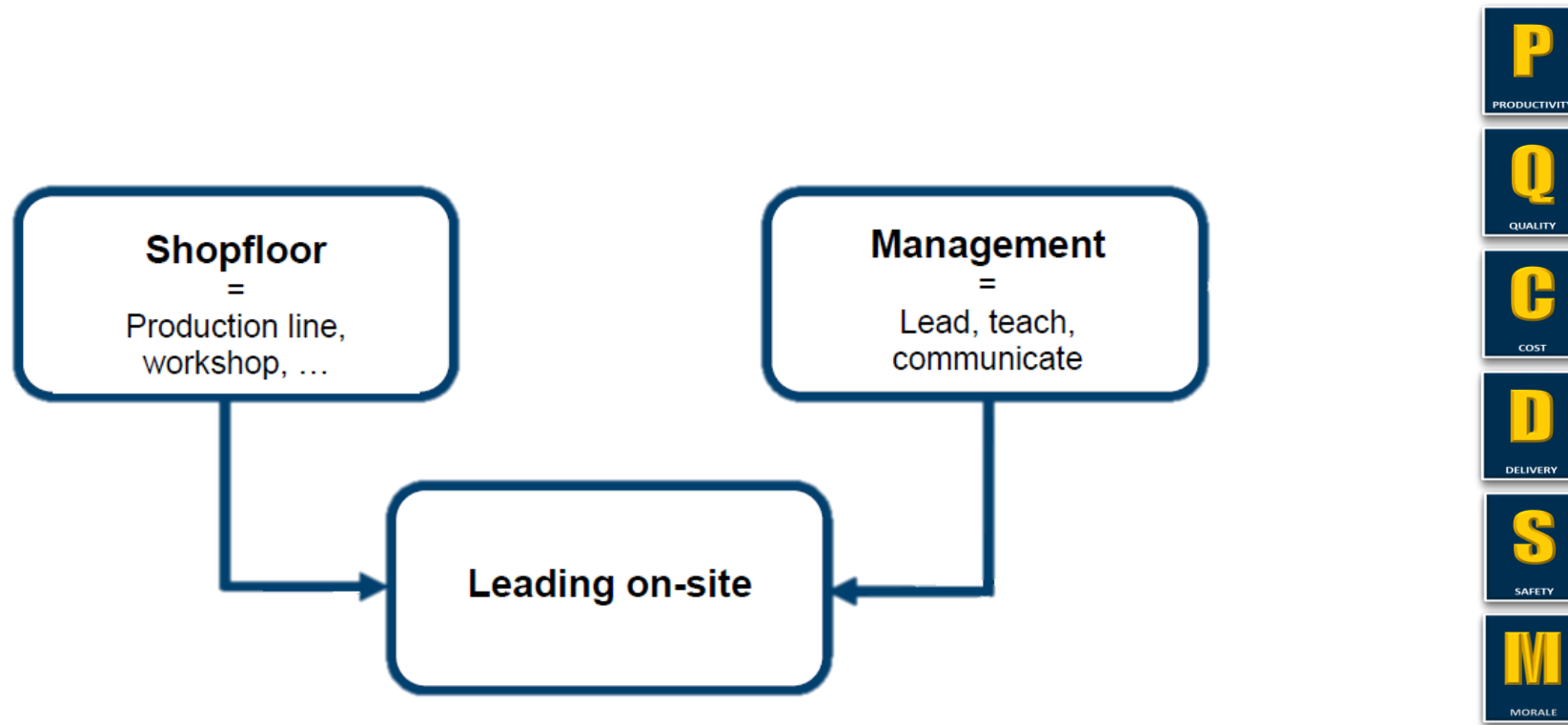
Where it fits?

The essence of Shop floor Management has four components:



What is Shop Floor Management?

SFM are the activities systematically done into the Shop Floor Operation area in order to achieve the Operation KPIs of Safety, Quality, Delivery, Cost, Productivity and Morale.



Why do it?

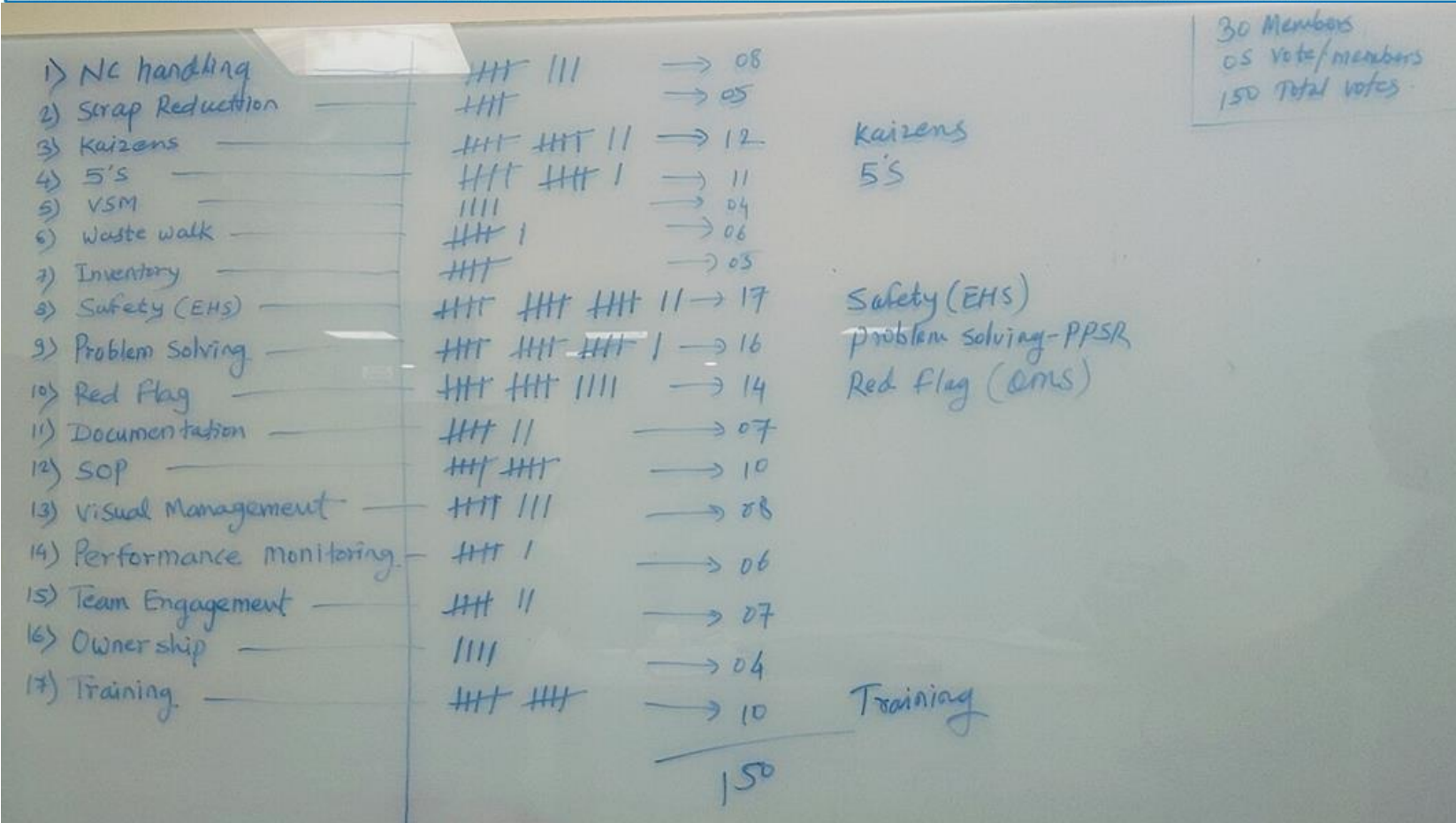
Rationale Behind Shop Floor Management



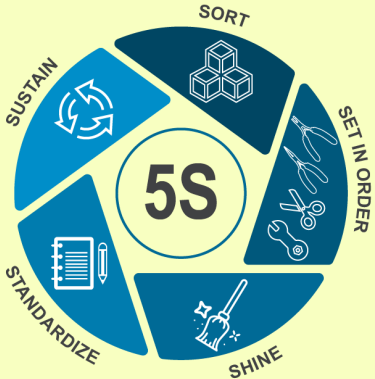
Brainstorming & multi voting for SFM topics



Selection of Shop Floor KPIs



SFM Agenda (ACT)– Awareness, Coaching and Training

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
				
Verification and give awareness to team members:- ➤ Importance of PPE, ➤ Emergency Assembly Point, ➤ Emergency contact number, ➤ Safety point understanding and verification in TPM, ➤ Spillage of Oil, Coolant ect, ➤ Use of ESD & importance, ➤ Gangway free and not blockage, ➤ Fire extinguisher are easy to access and not blockage, ➤ Electrical hazard, ➤ Any unsafe condition at line.	Weekly assessment. ➤ Understanding about 5s, ➤ Understanding about colour coding standardization, ➤ Both assessor and line will be selected randomly- Software, ➤ Average of weekly score is taken for Monthly score, ➤ Assessment process is digitized. Monthly appreciation-Best 5S Zone of the Month Trophy.	Verification and give awareness to team members:- ➤ Handling of NC parts on line. ➤ Verification of supermarket trolley, parts identification. ➤ Possibility of part mix up and verification of detection method. ➤ Line documentation verification. ➤ WIP part handling and ensure not exceeding capacity. ➤ Contamination audit at station	Weekly review of Live PPSR's ➤ Each PPSR leaders presenting his/her project, ➤ Right use of problem solving tools. ➤ Completion of each stage of PPSR and coaching by Coach. ➤ Possibility of horizontal deployment of results. ➤ Best PPSR's are selected for external competitions. Monthly appreciation-Best PPSR of Minifactory / Department- Appreciation Certificate & Cash prize.	Team visit-Kaizen Tree, collect all suggestion. ➤ Review and select Kaizens. ➤ CFT visit at location for review feasibility of implementation, ➤ Defining dedicated team for implementation by MF, ➤ CFT evaluate effectiveness of Kaizen- in terms of Safety, Quality, Productivity, Cost, Delivery, & Moral. Monthly appreciation-Best Kaizen of Minifactory / Department- Appreciation Certificate & Cash prize.

SFM Agenda:-

MONDAY



Safety Stewart Team, Takwe

										
Pump Line 1 Tushar Kamathe	Pump Line 2 Chandrasekhar	Pump Line 3 Dinesh Kuldharan	Pump Line 4 Sagar Fadtare	Actuator Line 1 Sanjay Shende	EMEGR Line 1 Rajendra Banpure	Module Line - 1 Sagar Borade	Actuator Line 2 Raju Dhar Kundu	EMEGR Line 2 Rahul Patil	Module Line 2 Ajay Jambhulkar	PEGR Line Sanjay Gupta
										
Machining Sunil Pradhan	Machining Rajendra Jena	Machining Abhijit Chavan	Machining Jayvant Patil	ETB Line Sudhir Shinde	BPV Line Amol Malpote	EAM+GEN 4+ Suresh Kute	SEGR Line Vishwajeet Patil	200% firewall area Ankush Kudale	NCM area Yogesh Gulave	

Roles & Responsibilities

Frequency

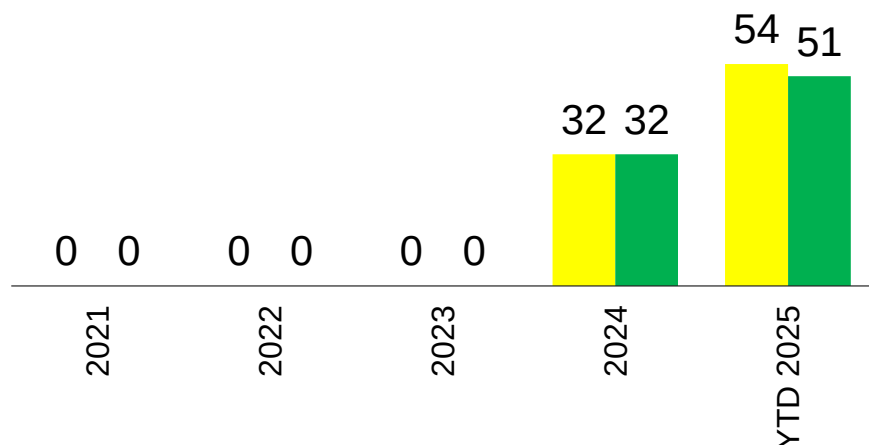
- | | |
|--|---------------|
| ✓ Safety Walk at respective lines | Daily |
| ✓ Escalation/ Information of Unsafe Act & Conditions in SFM Meeting | Daily |
| ✓ Safety meeting with Line OEs – Do's & Don'ts and Lessens learned @ Incidents | Weekly |
| ✓ Safety audits along with Safety Officer or Safety Committee Members | Weekly |
| ✓ Safety Kaizen | As applicable |



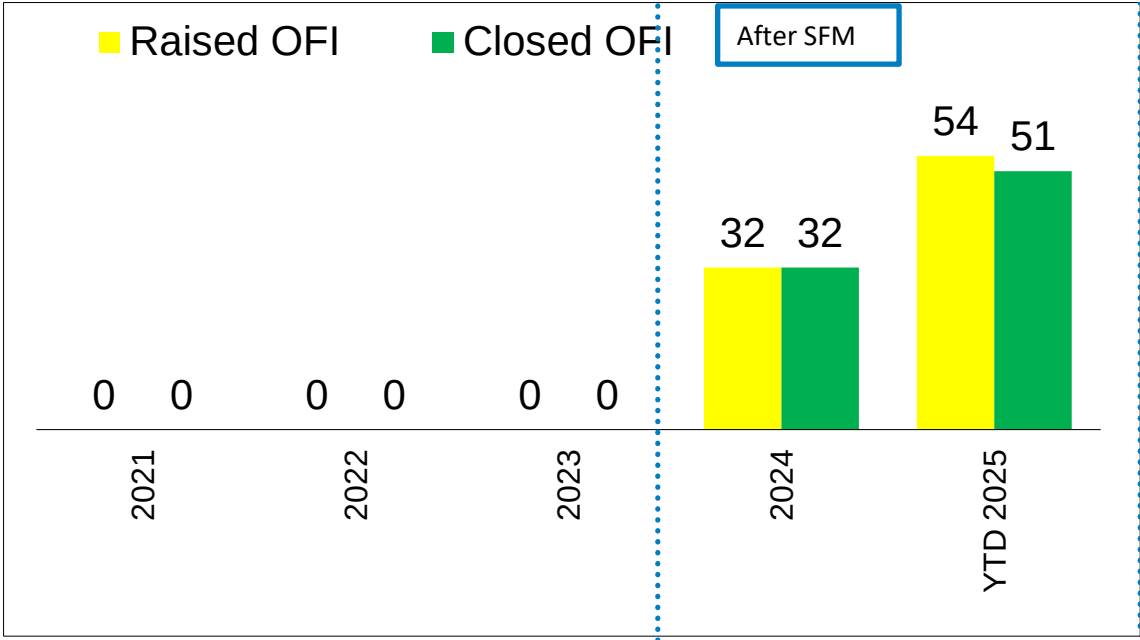
EHS OBESERVATION TRACKING SHEET

Sr no	Date	Shift	Raised by	Line	Before condition	Root cause	PCA	Responsibility	Target date	Status
18	25-Mar-25	A	Rajendra Jena	AMS1,AMS2	Air connection valve lever not available	Missing from connection	Replaced	Amol Patil	25-Mar-25	Closed
19	25-Mar-25	A	Rajendra Banpure	EMEGR-1	Chance of finger trap in clamp	No remain gap between clamp & the door	Bolt fitted.	Kanchan Patil	2-Apr-25	Closed
20	31-Mar-25	A	Amey	NX2	Earthing wire not connected on suction pump	Connected at wrong place	Connection corrected.	Amol Patil	31-Mar-25	Closed
21	8-Apr-25	A	Santosh	Line-3	Switch board in damage condition	Wear & tear	1/Board near FW station not in use so removed. 2/Board off switch replaced.	Kanchan Patil	8-Apr-25	Closed
22	20-Apr-25	A	Amey	Near Ford-2	Emergency door walkway floor in damage condition			Amol	31-May-25	Closed
23	20-May-25	A	Suresh Kute	EAM	St 05 After reset machine door going home position	Stopper get slip	Cylinder setting done	Shashank	27-May-25	Closed
24	27-May-25	A	Ajay Jambhulkar	Mod-2	HMI screen height not ok chance to hit head of operator	Wrong position HMI installed	HMI position changed .	Kanchan	10-Jun-25	Closed
25	10-Jun-25	A	Rajendra Jena		Hydro bull movement speed controller not working	Speed controller malfunctioning	Speed controller replaced	Nikhil	30-Jun-25	Closed
26	16-Jun-25	A	Rajendra Jena	Ford-1	Coolant chiller unit Motor cover missing	Cover broken	Cover replaced	Kanchan	16-Jun-25	Closed
27	17-Jun-25	A	Sagar Fadtare	NCM area	At entrance Guard grouting bolt come out & the hole is not filled	Properly fitting not done	One bolt grouting properly but other two bolt grouting not done due to	Amol Patil	30-Jun-25	Closed
28	24-Jun-25	A	Sagar Fadtare	Mazak	Extension of channel coming in walk way	Make place for wrong location	Removed extended channel	Manoranjana	24-Jun-25	Closed
29	24-Jun-25	A	Sagar Fadtare	Line-4	Tata TGD1 model fixture trolley in damage condition	Continue use	Wheel changed	Nagesh	15-Jul-25	Closed
30	24-Jun-25	A	Rajendra Jena	ETB			AS per CFT decision point is closed	Anand Mishra		Closed
31	8-Jul-25	A	Sagar Fadtare	Module-2	The location needs to be redefined	Wrong location done	Instead of two location made single location	Gunaji Parab	9-Jul-25	Closed
32	8-Jul-25	A	Rajendra Jena	AMS3	Main power supply cable position need to be changed		Cable routine done	Amol Patil	15-Jul-25	Closed
33	8-Jul-25	A	Santosh Wankhed	Machining area			Earthing connection done	Amol Patil	15-Jul-25	Closed
34	9-Jul-25	A	Amey Kanjale	DPS				Nikhil Patrike		Open
35	11-Jul-25	A	Amey Kanjale	Line-2 st 70	Fitting get loose and come outside		Cover fitted properly	Ramesh	12-Jul-25	Closed
36	14-Jul-25	A	Raju Kundu	Line-1	Cover missing.	Cover broken	New cover fitted.	Kanchan Patil	27-Jul-25	Closed
37	21-Jul-25	A	Rajendra Jena	Ford-1	Drum filter Jam	Drum filter not clean by nozzle.	Drum filter nozzle pipe cleaning checkpoint needs to be added to the preventive maintenance (PM) schedule.	Nikhil Patrike	30-Jul-25	Open
38	18-Jul-25	A	Rajendra Jena	Insight washing m/c FW tab	Switch broken	Continue use	Switch replaced still wire routine pending.	Amol Patil	27-Jul-25	Open

■ Raised OFI ■ Closed OFI

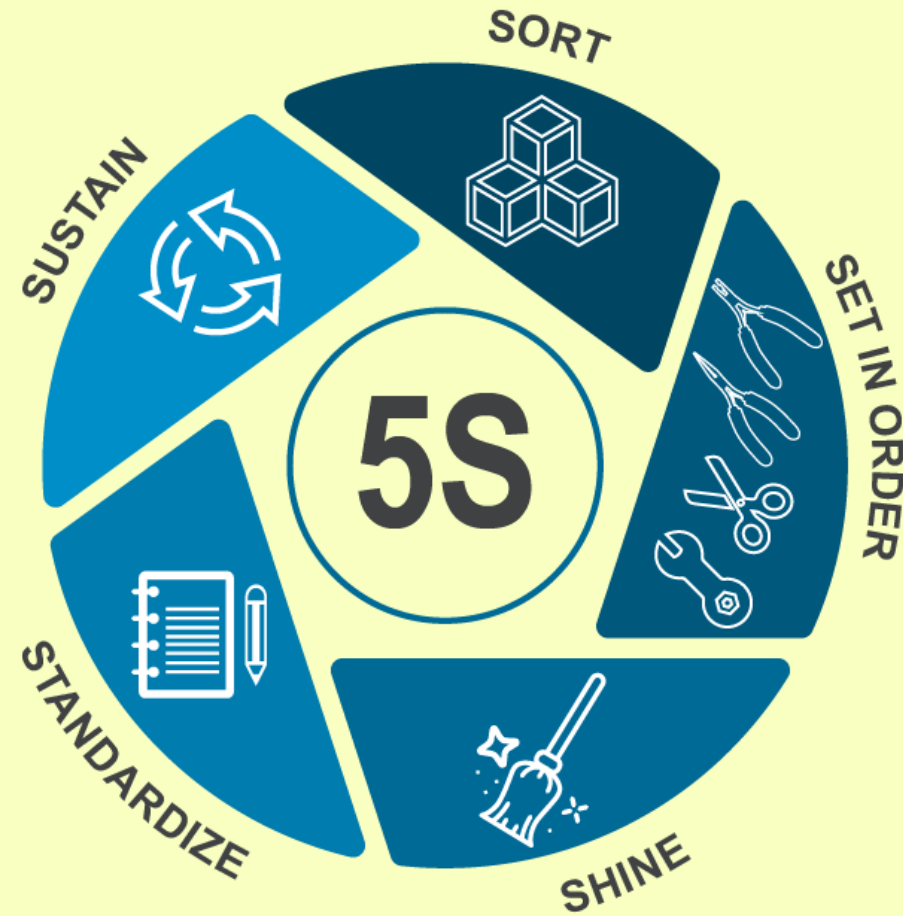


Pune Plant Safety, Health & Environment Summary

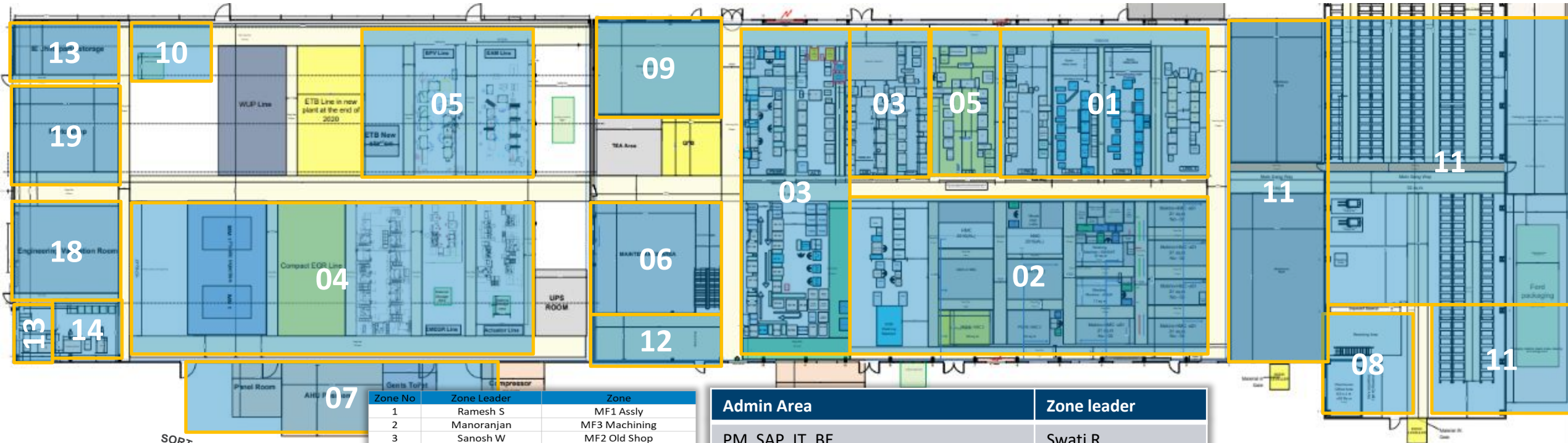


SFM Agenda:-

TUESDAY



5s Zone and Zonal Leader, Takwe



Zone No	Zone Leader	Zone
1	Ramesh S	MF1 Assly
2	Manoranjana	MF3 Machining
3	Sanosh W	MF2 Old Shop
4	Parab Gunaji	MF2 New shop
5	Ajay A	MF3
6	Angamuthu Kathiravan	Maintainance
7	Nikhil P	Utility & Outside plant
8	Vaibhav J	QA Incoming
9	Siddhant Kadam	QA Lab
10	Sanjay A	QA Customer
11	Siddu H	Logistics
12	Nitish Kulkarni	Industrial Engineering
13	Nitish Kulkarni	IE Cage and Machine building
14	Ramesh S	Oil test Area
15	Anil K	Office
16	Nikhil P	Outside Area
17	Shashikant Kore	Scrap Yard
18	Rishi Y	Engineering-PG
19	Ramachandra Shewale	Engineering-Pumps
	Santhosh PV	PLANT LEVEL

Admin Area	Zone leader
PM, SAP, IT, BE	Swati R
Sales	Monika
Front Office, Conference rooms, Visitors Rooms	Sonali Kumbhar
Engineering	Gaurav A
HR, MD Offices etc	Selina Ghatge
Purchase & SQD	Rajkumar Salgar
FICO	Deepak Lambe

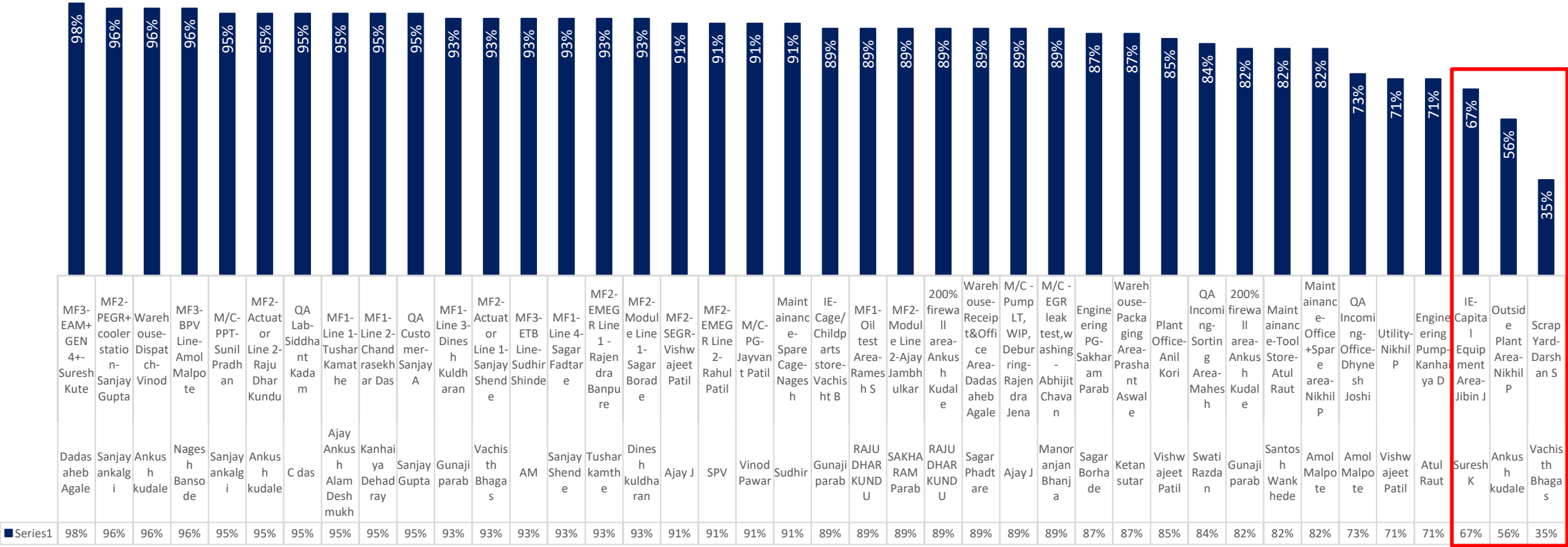
5S Evaluation

5S Evaluation	
Score Criteria	5-Available and ready for assessment, 4-Available but not accessible for assessment 3-More than three observations, 2-More than five observations 1-Not available
Question 1	Does the line or area you visited have the updated version of the 5S standard is available?
Question 2	Does the line or area you visited have the 5S Zonal/Line Leader Photo available?
Question 3	Do you see any dropped child parts, piece of paper, barcode sticker, cotton, packaging materials, hand gloves, ESD aprons, etc.?
Question 4	Is there a "Place for Everything & Every thing in Place"- assess with area 5S Standard if needed.
Question 5	Are the raw material pallets marked within BLUE floor marking?
Question 6	Are the FG pallet(s) marked within GREEN floor marking?
Question 7	Are the Empty bin pallets marked within ORANGE floor marking?
Question 8	Are the Stations, Cleaning stations, Tool Trolleys, and other items identified within WHITE floor markings?
Question 9	Are the Scrap materials, Dustbins marked with a RED floor marking Or scrap parts are In Red bins.
Question 10	Is there an identification label on every file, station, machine, drawer, storage shelf, etc.?
Question 11	Does the region, line or station have no debris, such as burrs, chips or dust?
Question 12	Please list three areas where you think we can do better.

5S SCORE



Week-16 (2025)





5S Audit Score - 2024																									LEAN Six SIGMA		
Zonal Leader		Ramesh S	Manoranjan	Santosh W	Parab Gunaji	Ajay A	Kathir	Nikhil P	Vaibhav J	Siddhant Kadam	Sanjay A	Vinod	Dadasaheb Agale	Prashant Aswale	Yogesh Gulave	Jibin J	Vaishant B	Ramesh S	Anil K	Nikhil P	Dashan S	Kanhaiya D	Saktham Parab	Santhosh P V			
7	July-24	92.6	91.6	90.1	91.7	93.6	86.0	91.0	64.0	89.0	92.0	89.0	73.0	90.0	90.0	89.0	76.0	93.0	90.0	92.0	93.0	78.0	87.0	87.3			
8	August-24	93.2	91.6	90.9	91.5	93.0	86.0	92.0	64.0	89.0	92.5	89.0	73.0	82.7	82.7	89.0	76.0	93.0	92.0	91.5	92.5	78.0	87.0	86.8			
9	September-24	92.9	92.4	92.1	92.5	93.5	86.0	92.0	64.0	89.0	92.5	89.0	73.0	82.7	82.7	89.0	76.0	93.0	92.0	91.5	92.5	78.0	87.0	87.0			
10	October-24	92.8	93.3	93.3	92.1	91.3	85.3	70.4	72.0	91.8	90.3	93.3	85.4	90.9	89.1	66.1	80.9	88.7	86.4	66.4	45.5	78.7	84.8	83.1			
11	November-24	91.2	91.5	92.0	92.7	93.1	88.1	76.4	76.5	95.5	94.5	92.7	91.8	87.3	89.7	70.9	69.1	93.6	78.2	78.2	40.9	0.0	96.4	80.9			
12	December-24	93.3	93.0	90.9	93.7	92.9	90.3	73.3	80.9	90.9	92.7	96.4	94.5	91.8	90.9	82.2	92.7	82.4	87.3	85.0	63.6	89.1	90.0	88.1			

ZONE-Warehouse-Dispatch
Zonal leader-Vinod Pawar

Best 5s Zone of the Monthly



5s BE Assessment result, Takwe, Pune

Standardization		Topic	0	1	2	3	4	5	Score																																																																													
1	Are 5S division standards method, process and templates applied and updated and are 5S stakeholders responsibilities defined?	5s	no method	5S method, process and templates following the division standard are applied on the shop floor. 5S responsibilities are defined but not standardized across the organization. No 5S zones are defined.		5S method, process and templates following the division standard are applied in all following areas (shop floor/office floor, meeting rooms, offices, etc.). 5s included in the planning for new lines/areas, offices. In production: New lines are fully integrated in 5s at SOP. 5S development is under responsibility of BE / 5S coordinator and/or team leader. All 5S zones are defined		5S method, process and templates following the division standard are applied in all areas (shop floor/office floor, meeting rooms, offices, file server, outside area, etc.) and updated if required. 5s included in the planning for new lines/areas, offices. In production: New lines are fully integrated in 5s at SOP. Management 5S responsibilities (e.g.. Zone responsibility, part of escalation step, member of steering committee, etc.) are clearly defined, communicated and standardized across the organization.	5																																																																													
2	What is the level of implementation of 5S in all areas (shop floor/office floor, meeting rooms, offices, file server, outside area, etc.) depending on assessed area?	5s	Distribution level <table><tr><td>100</td><td>0</td><td>2</td><td>3</td><td>4</td><td>4</td><td>5</td></tr><tr><td>90</td><td>0</td><td>2</td><td>3</td><td>4</td><td>4</td><td>4</td></tr><tr><td>80</td><td>0</td><td>2</td><td>2</td><td>3</td><td>4</td><td>4</td></tr><tr><td>70</td><td>0</td><td>2</td><td>2</td><td>3</td><td>3</td><td>4</td></tr><tr><td>60</td><td>0</td><td>1</td><td>2</td><td>2</td><td>3</td><td>3</td></tr><tr><td>50</td><td>0</td><td>1</td><td>2</td><td>2</td><td>2</td><td>3</td></tr><tr><td>40</td><td>0</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td></tr><tr><td>30</td><td>0</td><td>1</td><td>1</td><td>1</td><td>2</td><td>2</td></tr><tr><td>20</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>10</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table> Application level 01s2s3s4s5s						100	0	2	3	4	4	5	90	0	2	3	4	4	4	80	0	2	2	3	4	4	70	0	2	2	3	3	4	60	0	1	2	2	3	3	50	0	1	2	2	2	3	40	0	1	1	2	2	2	30	0	1	1	1	2	2	20	0	1	1	1	1	1	10	0	0	1	1	1	1	0	0	0	0	0	0	0	5
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3	Does the organization check the 5S program effectiveness and is the program effective? (for areas where 5S is fully implemented)	5s	no activities in place	Defined 5S audits are conducted on regular basis. KPI's are updated. 5s is visible on the shop floor. The last audit results (min. 2) show positive trend or if already on high level (>95%) the results are sustained. SMART targets for the 5s audits are defined for one year.		5S audits are planned and conducted on a regular basis. KPI's are communicated across organization. Improvement actions are defined, followed. Action plans are up to date. 5s is visible in following areas (shop floor/office floor, meeting rooms, offices). The last audit results (min. 2) show positive trend or if already on high level (>95%) the results are sustained. SMART targets for the 5s audits are defined for one year for each zone and were adjusted if achieved earlier. 5s program is continuously improving (pro-active approach).		Counter measures between audits are established to ensure efficiency (e.g.. Daily walks, random checks, layered audits etc.). Actions are defined, worked on and closed in time. 5s is visible in all areas (shop floor/office floor, meeting rooms, offices, file server, outside area, etc.). The last audit results (min. 2) show positive trend or if already stabilized on high level (>95%-100%) the results are sustained. SMART targets for the 5s audits are defined for one year for each zone and were adjusted if achieved earlier below 95%. 5s program is continuously improving (pro-active approach).	5																																																																													

100% score achieved during 2024 BE assessment

SFM Agenda:-

WEDNESDAY





Verification and give awareness to team members:-

- Handling of NC parts on line.
- Verification of supermarket trolley, parts identification.
- Possibility of part mix up and verification of detection method.
- Line documentation verification.
- WIP part handling and ensure not exceeding capacity.
- Contamination audit at station

What we do in QMS?

☐ Planed Audits

☐ Red flag Monitoring with Actions

☐ Monitoring in-house quality incidents & Actions

☐ Reword recognition for best caught



DQS audit

Audit conducted on –
 May 2025
 Closed on –
 June 2025

Actions closed



Process audit

As planned audits
 conducted in June
 2025 & Next will be
 plan in Aug 2025

Actions Completed



Product audit

78 product audits
 conducted in 2024 vs
 planned 90

Due to No Production
 plan /low volumes
 audits are
 rescheduled



Layered audit

6 Layered process
 audits conducted Vs
 planned / required
 11nos.

Plan circulated
 New method adopted
 of spinning wheel

Will be completed



Customer audit

5 Customer audit
 conducted in 2025

Actions of FIAT audit
 are in progress.

Layer process Audit

Layered process audit – Spinning wheel



Auditor – Gaurav D
Area after spin -
Machining -Makino



Auditor – Ganesh Hole
Area after spin -
Actuator line 2



Auditor – Rana Singh
Area after spin -
Quarantine area



Auditor – Kathiravan
Area after spin -
ENEGR line 2



Auditor – Vashishth B
Area after spin -
Module line 2



Auditor – Simanta Sahoo
Area after spin -
Warehouse FG

LPA Audit
plan



Layered Process audit plan - Spinning wheel : 2024																									
Auditor Name	Department	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24
Anil Mahapatra	Production																								
Sudhir Sarda	Production																								
Chandrasekhar Das	Production																								
Ganesh Hole	OD																								
Shashank A	Man.																								
Bhushan Jadhav	E																								
Suresh Varshede	Production																								
Shashank Das	Quality																								
Rishi Kumar	E																								
Yashwantrao	Quality																								
Vishal Jagde	Quality																								
Rohan Anand	PPM																								
Rohan Kumar	QMS/AG																								
Gaurav Daga	PPM																								
Mahendra Patil	Quality																								
Nitin Datta	E																								
Vishal Chougale	Man.																								
Mangesh Jadhav	E																								
Saurabh	BE																								
Suresh Varshede	ECM																								
Allen Jose	E																								
Srinivas	SOD																								
Srinivas	Logistics																								
Anil Patil	Utility & Facility																								
Nitin Patil	Utility & Facility																								
Sanjay Sarda	E																								
Pramod	Logistics																								
Kundan Gunde	Logistics																								
Ranjith Raju	PH																								
Sanjay P V	Production																								

LPA Audit
check sheet



KSPG Layered Process Audit - Assembly line			
Product	Line No.	Date	Rev. No.
Famb	Line-3	17/08/25	01
Auditor	Raja Shree Kumar		Rev. Date: 04/09/2025
Week			Page No. 1 of 1
Process	Activity to be evaluated	Score	Observation
Safety	Safety sensor / switches are working. Guards are in place?	5	OK
5S & Identification	Does the order and cleaning of the machine comply with the 5S standard?	5	OK
Leak Test	Check gas / FGD identification done on bins / containers?	5	OK
Screwing	Are the screwdrivers correctly functioning and calibrated as per defect?	5	OK
Pressing	Are the stationing stations functioning correctly and the load thresholds set are consistent with what is stated in the specification?	5	OK
Lubrication	Are the supply stations functioning correctly and the process thresholds set are consistent with what is specified in the specification?	5	OK
Functional Test	Are the functional test stations functioning correctly and the process thresholds set are consistent with what is specified in the specification?	5	OK
Free Rotation	Are the process controls carried out as per control plan?	5	OK
Anti Error system (Poka Yoke)	The anti-error systems are correctly functioning	05	OK
Vision System (Camera check)	Are vision systems functioning correctly?	05	OK
Marking and traceability of the finished product	Is the marking station working properly?	05	OK
Management of non-compliant product	Is the finished product correctly marked and therefore consistent with the provisions for traceability?	05	OK
	Is the management of the non-compliant product consistent with process of handling non conforming parts?	05	OK
	Is the part correctly identified and segregated?	05	OK

LPA Audit open point closing action list

					Responsible:		Pune - S & A division	
					Author:		Mohini Karnik	
0% 10% 20% 30% 40% 50% 60% 70% 80% 90%								
Date	Customer	Product name	Audit type - Customer Internal	Auditor team	Audit observation	Root cause / Probable root cause	Action	Type of Issue
1/9/2024		Line 3 Pump	Internal - Layered process	Mr. Tapas Kumar	Quality issue- Operator is unaware about customer complaint.	20012024-No customer complaint booked till date.	20012024-Awareness given to OE-how to face audit	Discipline
01.03.2024		ETB	Internal - Layered process	Mr. Sanjay shende	overall SS not maintained on station	Discipline not follow	Awareness given to all employees	
01.03.2024		Line 2	Internal - Layered process	Mr. Ajay alam	One point lesson not available on line	15032024-All One point lessons are told to remove from line-as per QA	15032024-New guidelines to be deployed- Resp- Process QA	System
01.03.2024		Line 1	Internal - Layered process	Mr. Jibin jose	OP 10, 50 child part bin location not defined, it can fall	15032024-High chance for child part fallen down-need engineered solution	Kaizen done- Attach reference	
1/7/2024		Line 3 Pump	Internal - Layered process	Mr. Tapas Kumar	St 95- Oil qty not defined in CP	11012024-As per process-qty is not need to be defined - only oil splash is defined-for free rotor movement	11012024-No actions required	
01.03.2024		Line 1	Internal - Layered process	Mr. Jibin jose	OP 50- oil quantity requirement on inspection report is not same as mentioned in PP/CP, lower limit not available	15.03.2024-Lab used wrong model report-BR10 model report(print out) used for F&B	15.03.2024-Awareness given-Resp-Sohan K	
01.03.2024		Line 1	Internal - Layered process	Mr. Jibin jose	fire all person is not identifiable based on skill matrix	15032024-As per guideline, FW person names is available skill matrix.	15032024-Double verified Mr. Sachin Tadva was doing FW, as per system he is eligible for FW, same found in skill matrix as well	Discipline
02.03.2024		Line 4	Internal - Layered process	Mr. Santosh W	NC part kept at non identified area	07.04.2024-This location was created temporary storage of Dry test NCK pump-which was waiting for oil test bench availability.	07.04.2024-As per CFT, this location is shifted to Oil test area-Closed	System
02.03.2024		Line 4	Internal - Layered process	Mr. Santosh W	St 30- DSM display not readable	07.03.2024-CFT visited station-found tube light is getting covered DSM screen, need to		
05.04.2024		Hyundai 1.5	Internal - Layered process	shashank A	op 60- EGR bin identification not available	07.04.2024-This bin was created temporary storage for keeping EGR due to Std. bin given for washing	Spare std. bin set when regular bin send for washing	System
05.04.2024		Hyundai 1.5	Internal - Layered process	shashank A	St 70 half screwing work instruction not available	As this is NG part, no need for WI. In defect matrix detail information available for half screwing part	How to handle half screwing part information updated in work instruction	System
05.04.2024		Hyundai 1.5	Internal - Layered process	shashank A	skill matrix to be updated	15.04.2024-As per guideline, FW person names is available skill matrix.	15.04.2024-Double verified Mr. Vismamoni was doing FW, as per system he is eligible for FW, same found in skill matrix as	Discipline
05.04.2024		Hyundai 1.5	Internal - Layered process	shashank A	St 30 PP not available on line	During rechecking found PPE matrix available on line itself	Reverification also found PPE available on line	Discipline
14.03.2024		Line 1-3D15	Internal - Layered process	Mr. Sanket S	Customer complaint- not displayed on line. Line operator is unaware about CC	21032024-No customer complaint booked till date.	21032024-Awareness given to OE-how to face audit	Discipline
05.04.2024		Hyundai 1.5	Internal - Layered process	shashank A	Op 30, 50- burr observed op 60, 70 oil observed	Found after screwing generated burr partial on station	Awareness given to OE - how to clean burr after hilly interval with use of vacuum cleaner	Discipline
13.04.2024		EAM	Internal - Layered process	Ramesh savant	FGI child part identification was not followed	Discipline not follow	Awareness given to all employees	
13.04.2024		EAM	Internal - Layered process	Ramesh savant	Packaging work instruction not available on line	Internal part so no need for packaging instruction		

Red flag Monitoring with Actions

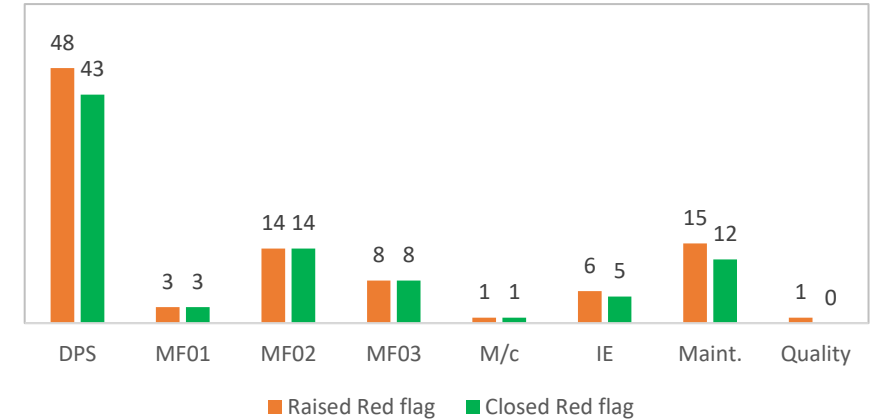
Red flag- RF-MF-2-12
Plant Quality to: Gunaji Parab, Santosh Wankhede, Yogesh Gulve
 Cc: Ramesh Sawant, Sandeep chaudhari, Rupesh Kanekar

02/06/2025 04:30 PM [Show Details](#)

RHEINMETALL	RED FLAG	Form No. FFQAR2
		Rev No. 00
		Rev Date 12/2/2019
Date 05.02.2025	Red flag- RF-MF-2-12	
Red Flag description (Non Conforming)	Activator-2 St-105 distance after bearing pressing specification - min - 15.45, Max - 16.17 but in FOI filed it 15.43	
Red Flag type	Defined system/documentation not followed Defined process not followed Defined Product parameter Not ok (not caught/detected in by defined system/machine)	
Operator Name	Aniket shinde	
Team Leader	Rajendra bapure	
Reference Picture		

RED FLAG CORRECTIVE ACTION PLAN & CLOSURE			
Correction / Containment	Awareness given to the team member & FPI report reverify		
Root Cause.	Height is checked by vernier ad due to this chances of variation		
Corrective Action Taken	New flash pin gauge implimented		
Documents Updation	Documents	Applicable	Done
	PFD	NA	
	PFMEA	NA	
	Control plan	A	Done
	work instruction	NA	
	Process Parameter	NA	
	TPM / Pokayoke Check list	NA	
	First / Last piece inspection	A	Done
	Master verification sheet	NA	
	PM Check list / Plan	NA	
Effectiveness Verification	Problem Reoccured ?	Y/N	No
	Corrective actions followed ?	Y/N	Yes
Closure Status	Closed / Open	Closed	

Raised Vs Closed Red flag Status 2025



Red flag tracking sheet

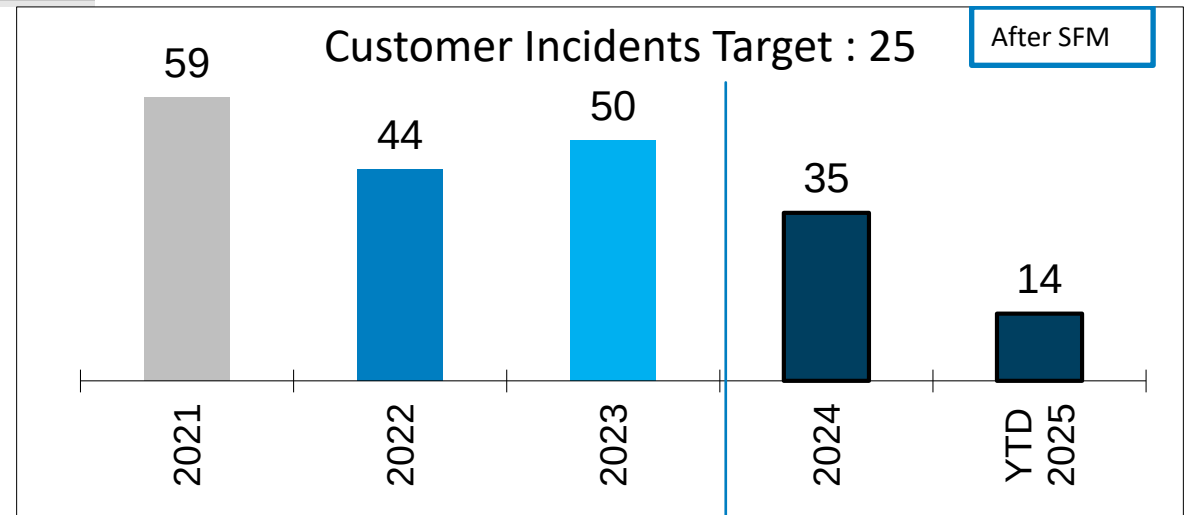
SR	Raised Date	MF nan	Line /Area name	Station name	Red Flag No (RF-MF-01)	Responsible Department	Red Flag description	Evidence (Photo or File)	Team leader	Immediate Action taken	Action taken B	Immediate Action taken D	Systematic permanent action	Target date	Closure responsibility	Status
1	7-Jan-25	MF2	EMEGR-2	ST-80	RF-MF-01	Production	During line audit in Emegr line -2 it is observed that barcode pasted in fixture to bypass Molycoat gun tip sensor .		Rajendra Bapure	1)Removed barcode & awarness given to operator	Gunaji	7-Jan-25	1)Shift end meeting given awarness to all new team leader to any found barcode paste on fixture please inform to supervisor don't run line till it correct.	7-Jan-25	Gunaji	Closed
2	14-Jan-25	MF3	ETB	ST-40	RF-MF-02	Production	During line audit in Etb line, it is observed that sticker pasted in fixture at op 40 Gap checking station .		Sudhir Shinde	1)Removed barcode & awarness given to operator	Ajay A	14-Jan-25	1)Shift end meeting given awarness to all new team leader to any found barcode paste on fixture please inform to supervisor don't run line till it correct.	14-Jan-25	Ajay A	Closed
4	18-Jan-25	MF1	Line- 3	St-100	RF-MF-04	IE	Line-3 E511 St 100 Plug Screwing specification mismatch found in system. Plug screwing specification given torque 7-8 Nm, Angle 0-120 instead of 28-30 Nm, Angle 0-100		Vashistha	PLC correction done	Vashistha	23-Jan-25	All relevant models/Station to be verify for similar observations - Prod. Team leader	24-Feb-25	Vashishtha	Closed
5	22-Jan-25	MF2	FW area	200% firewall	RF-MF2-05	Production	Loose contamination found on 200% firewall station .		Ajay Jambhulkar	1) awarness given	Gunaji	22.01.2025	2) 5S checksheet added for all FW station.	31-Jan-25	Gunaji	Closed
6	25-Jan-25	MF1	Line- 2	70B	RF-MF1-06	Maintenana ce	P VP ST 70 We observed during audit screwing gun calibration date over due.Nov 24		Chandrashekhar Das	Calibration done	Sidharth Shankar	27-Jan-25	Spare material calibration activity started from this year	5-Feb-25	Sidharth Shankar	Closed

Monitoring in-house Quality incidents & Actions

Date: 08.08.2025				Minifactory :-		MF1		Responsibility :-		Anand M		Other Topics	
Topic		Status	Line Name		OEE	Availability	Efficiency	Quality	Escalation	Remark			
EHS	Safety Logging Indicators	No	ASSEMBLY	Line1	84.09%	92.70	96.30	94.20		A- 3D15- Starting position nok- 90803- 15 mins. Q- 3D15- RR1-14 (R.ok), RR2- 11(R.ok)		3D15 Total= 458 Ok= 430	
	Safety Leading Indicators	No		Line2	92.63%	100.00	95.20	97.30				PVP Total= 219 Ok= 213	
	Environment Incidents	No		Line3	84.34%	92.70	93.80	97.00		A- E511- St40- Cycle not start- 90803- 65 mins. E- Setup change - 40 mins.- SMED Project- Ramesh S Q- 4D15- EV-6 (R.ok), Rotor press nok-2		4D15 Total= 828 Ok= 810	
Quality	Q Claim	No		Line4	No Prod.						No Plan		
	Q Claim	No											
	NC Cost	0.13											
Absenteeism	Total-Require	60	No's										
	Total- Availab	50	-10										
	Absent	6.00%	3										
	Leave	4.00%	2										
Cum Total				Cum Opes		Yesterday							
1	QMS Audit	0	0	0									
2	Red Flag	7	0	0									
3	In-house Quality Incident	5	5	0									
4	Supplier concerns on line	147	147	0									

Monitoring in-house Quality incidents & Actions

MF	Date	Imp	Product Name / Line	Problem Description	Root cause/ most probable cause identified	Action Plan	Target date	Responsibility	Action Status
MF2	3-May-25	IQI	Actuator assy-2	Due to motor housing sticker stuck inside motor gear Hyundai module fail for offset test	Sticker stuck inside motor housing while gear holding clamp movement at st160	LCA: 1.After barcode scanning at st160 Motor housing line barcode sticker location change(Remove barcode and paste on motor cover cap) - Closed . 2.OPL displayed with information of "All three barcode should be present with actuator at FW station" - Closed . PCA: 3.Motor housing line barcode label replaced with 8x13 mm barcode label and st160 fine tune for reading barcode (Not feasible to implement) - NA . 4.Feasibility to implementation of Fix location auto scanner with detection of part and correct location of barcode sticker - Closed .	05-05-2025 05-05-2025 15-05-2025 15-05-2025	Gunaji Rupesh Nityanand Kanchan	Closed Closed Open Open
MF3	12-May-25	IQI	ETB Line	EtB Panther - Half screw parts found at 200% FV - 6 nos	Cross screwing & LYDT detection malfunction on ST90	1. Cross screwing by new operator, training to new operators. - Closed . 2. LYDT setting checked & corrected. - Closed . 3. horizontal deployment of all other models to be check & variation of Renata & Ejot cover to be consider. - Closed .	16-May 16-May 15-Jun	Mangesh	Closed
MF2	20-May-25	IQI	EMEGR-1	Welding missing part found in FV	Manual barcode generate from st60	1. Manual barcode generate and print option disabled for line operator use. - Closed . 2. Dedicated rework station for EMEGR line 1 PFR raised under PO. - Open .	25-05-2025 30-07-2025	Mangesh	Open
MF2	23-May-25	IQI	Module Line 2	Hyundai module st70 water tube crack after screwing-1	- Plastic Tube over molding - Brass insert uls - Adaptor M6 hole position shift	1. Cavity 2 of Pipe separated, as all crack in cavity 2. 2. Pipe Brass insert checked & deformation found after screwing 3. Check point added at FV. - Closed . 4. Adapter M6 hole TPG go side not pass complete- Containment started with TPG - Ongoing Communicated to supplier - Closed . 5. under monitoring - Supplier 8D awaited. 6. Tube material missing grade composition inspection at LAB planned (30 Jun) under PO	4-Jun-25 10-Jun-25	Rupesh K	Closed
MF2	31-May-25	IQI	Module Line 2	St 60 screw broken-1-Defective screw found at st 60	Desouter unit fgun malfunction Thread depth uls or short	1. Desouter gun replaced with new gun, issue resolved. - Closed . 2. Problem of Desouter get to know from supplier. 3. Adapter M6 hole TPG go side not pass complete- Containment started with TPG - Ongoing Communicated to supplier - Closed . 4. Angle control not in working from 8 may 2025, control re-established with max turn angle control limit 100 degree - Closed 10/05/2025	31-May-25	Nikhil	Closed
MF1	4-Jun-25	IQI	Line 2	Panther VP with gap at FV	Gasket pinch (O ring) - manual placing	1. One point lesson to display & awareness training to operators. - Closed . 2. Concentration diagram & operator wise rejection monitoring of further failure to be recorded. - Open . 3. Camera Check image to be verify on ST95 for feasibility of Seal orientation. - Open .	30-Jun-25	Sohan Ramesh S Nilesh D	Open
MF1	5-Jun-25	IQI	Line 1	Fam B YOP vane mix up - 3D15 vane mixed	Vane mix up possibilities, at line fallen parts mix up and at dismantling area mix up material feed. -> Most probable suspect SS activity before Mr. Muller visit. (fallen vanes)	1. All FG stock of 180 nos rechecked by free rotation by manual method. Vane difference > 0.4mm, Pump gets locked fully. - Closed . 2. All model vane stock at Line 1 to be verified. - Closed . 3. Awareness training of how to handle fallen parts provided. - Closed . CFT meeting conducted & above possible improvement are listed, as no possibility to pass the concern part to customer, aspart will be jam at EOL bench, so no risk.	26-Jun-25	Sohan & Ramesh CFT	Closed
MF1	5-Jun-25	IQI	Line 3	4D15 YOP Shaft pressing with wrong orientation - 1nos	manul mistake - no poka yoke for orientation only shaft presence poka yoke suitable.	1. CFT revived this case on 26-Jun on GEMBA, this 1st case by new operator in last 3 years of production, so occurrence rate is very low. - Closed . 2. No risk of passing this part to customer from EOL & firewall inspection. - Closed . 3. No immediate risk & not a feasible, as earlier tool design was wrong this issue but risk of damaged shaft was less high.	26-Jun-25	Ramesh S Sohan & Sandeep	Closed

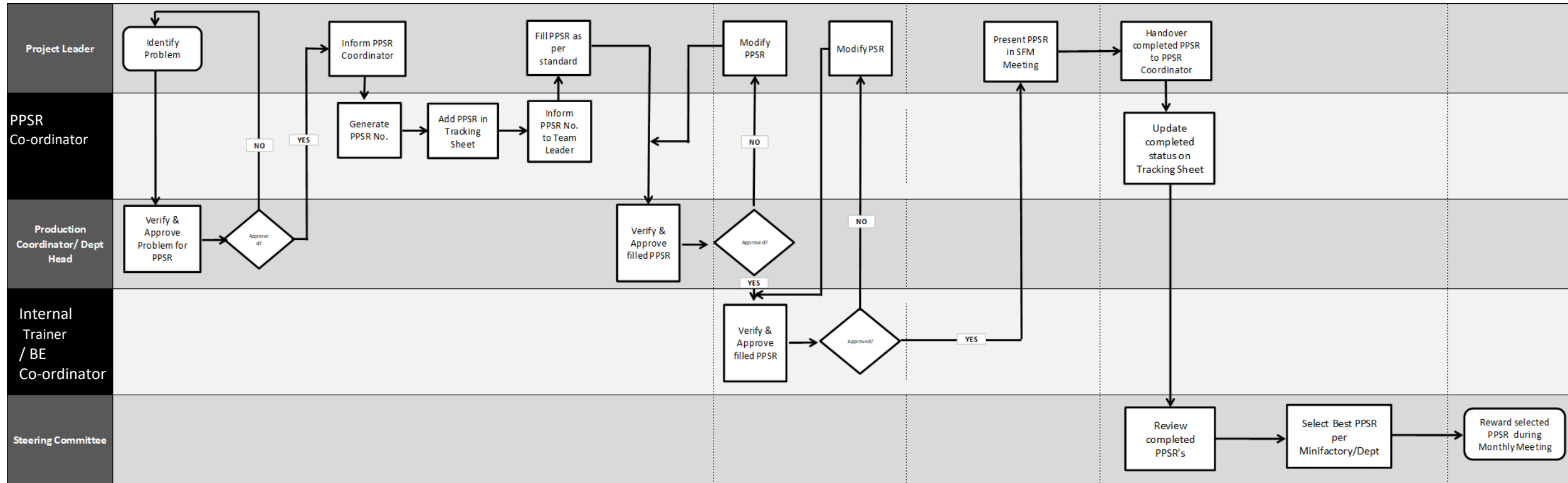


SFM Agenda:-

THURSDAY



Process map for the PPSR process



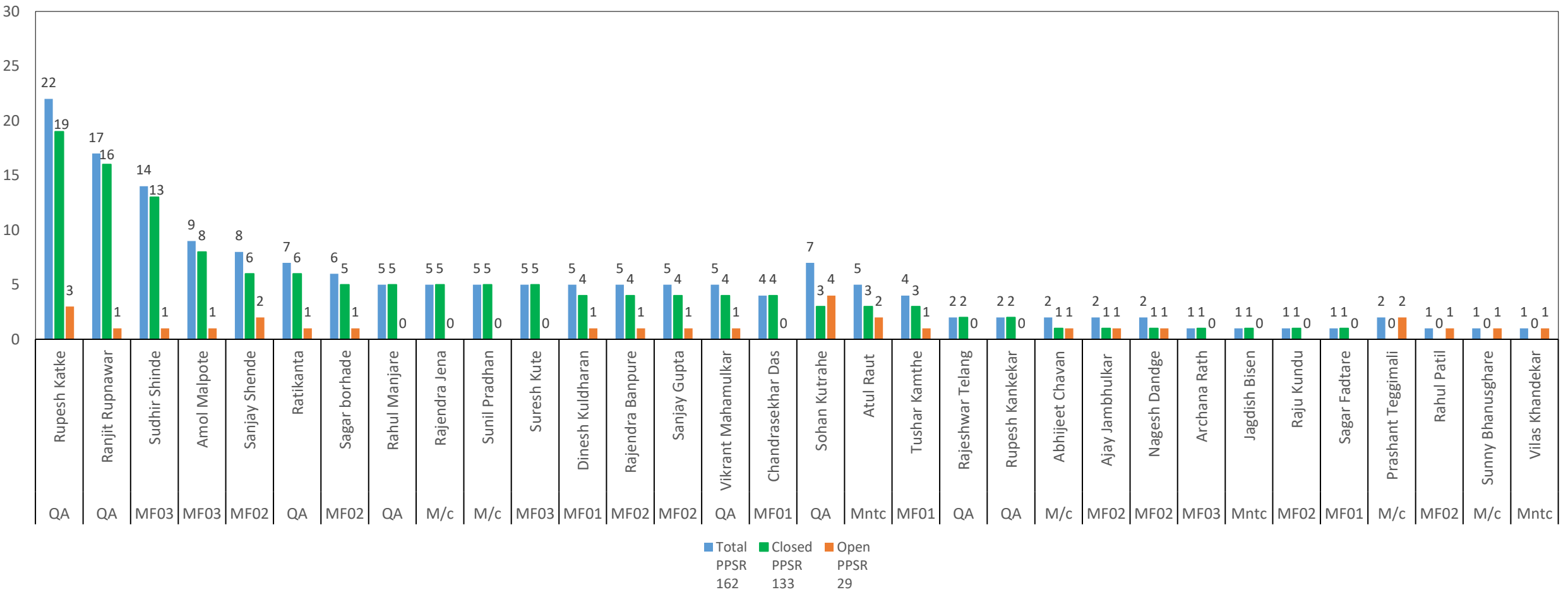
Note: Raising PPSR Terms & Conditions

- Confirmation Test is required to close PPSR
- If supplier concern, 8D report needs to be available till at least 6/8D. Report needs to be attached with PPSR

PPSR will remain open till Corrective action, standardization & HD is closed

- PPSR for breakdowns to be raised if Cause is unknown and defect is occurring more than 3 times in a month/ special drive from management. These can be led by team leaders, along with Maintenance Team member
 - PPSR for long and complex breakdowns lasting longer than 4 hours in a shift needs to be led by Maintenance team member
 - PPSR for FTQ to be raised as per Escalation Matrix. These can be raised by any CFT member
 - PPSR for performance to be raised if startup time is higher than 10minutes.
 - PPSR for performance to be raised if setup time is higher than defined standard
 - PPSR for Material issues for a line/product to be raised if there is material shortage for more than 2 hours in a shift & occurring more than 3 times in a week*. These can be raised by warehouse/Production Team member
 - PPSR to be raised if driven from management
- * Not applicable if on time production booking not done, SAP not working, Q/S hold

PPSR Summary 2024



34 Nos.PPSR's Uploaded in JIRA for year 2024

S.No.	Projects	Project Leaders	Department	Champion	Approach	JIRA	Jira ID	Estimated Cost Savings (Euro)	Status	Cost Saving Actual - INR	Cost Saving Actual - Euro	Impacted KPI	Scheduled Date of Project Update PPT
52	PPSR/2024/Jan/16 Hyundai Valve Hsg CMM Dimension & Relation Gauge Variation	Sunil Pradhan	Production	Anil Kori	PPSR	Yes		45000	Completed	4502664	48942	NCC	Closed
40	PPSR/2024/Apr/05 TATA 3.3L Hsg M/C Valve Hsg Unclean @Dia 10mm	Ranjit Rupnawar	Quality	Anil Kori	PPSR	Yes	PU2024-55	45000	Completed	4480078	48696.5	NCC	Closed
34	PPSR/2024/June/09 EAM Calibration NOK	Amol Malpote	Production	Anil Kori	PPSR	Yes	PU2024-49	45000	Completed	4444714	48312.1087	NCC	Closed
39	PPSR/2024/Jan/18 TATA 1.5L M/C Part Leakage NOK	Sunil Pradhan	Production	Anil Kori	PPSR	Yes	PU2024-54	45000	Completed	4243384	46123.73913	NCC	Closed
41	PPSR/2024/May/04 Actuator 1 Motor Housing Stuck up	Sanjay Shende	Production	Anil Kori	PPSR	Yes	PU2024-56	45000	Completed	4164408	45265.30435	NCC	Closed
35	PPSR/2024/Jan/17 M/C Makino Cap fitment distance variation	Sunil Pradhan	Production	Anil Kori	PPSR	Yes	PU2024-50	35000	Completed	3483852	37867.95652	NCC	Closed
63	PPSR/2024/May/02 EAM Spring Carrier & Shaft welding NOK	Ranjit Rupnawar	Quality	Anil Kori	PPSR	Yes		32000	Completed	3386704	36812	NCC	Closed
36	PPSR/2024/Feb/12 EAM Calibration Test NOK	Archana Rath	Production	Anil Kori	PPSR	Yes	PU2024-51	35000	Completed	3276806	35617.45652	NCC	Closed
37	PPSR/2024/Feb/07 VECV 494 Water Cycle Leakage	Rajendra Banpure	Production	Anil Kori	PPSR	Yes	PU2024-52	35000	Completed	3136993	34097.75	NCC	Closed
33	PPSR/2024/May/11 ETB Welding Incomplete	Amol Malpote	Production	Anil Kori	PPSR	Yes	PU2024-48	25000	Completed	2446829	26595.96739	NCC	Closed
58	PPSR/2024/Aug/06 Hyundai Adapter Part on Relation Gauge NOK	Rupesh Katke	Quality	Anil Kori	PPSR	Yes		25000	Completed	2304600	25050	NCC	Closed
42	PPSR/2024/Feb/11 Acuator 2 Connector Shell Pressing NOK	Sanjay Shende	Production	Anil Kori	PPSR	Yes		25000	Completed	2299540	24995	NCC	Closed
51	PPSR/2024/Jul/06 Hyundai Valve Hsg Steel Ring surafce roughness NOK	Ranjit Rupnawar	Quality	Anil Kori	PPSR	Yes		20000	Completed	2155100	23425	NCC	Closed
32	PPSR/2024/May/03 BPV Push over Time NOK	Amol Malpote	Production	Anil Kori	PPSR	Yes	PU2024-47	20000	Completed	1815466	19733.32609	NCC	Closed
44	PPSR/2024/Jan/07 Actuator 2 Axle bend after pressing	Rupesh Kankekar	Quality	Anil Kori	PPSR	Yes		15000	Completed	1483684	16127	NCC	Closed
31	PPSR/2024/Oct/14 4D15 Free Rotation NOK	Sohan Kumar Kutrahe	Quality	Anil Kori	PPSR	Yes	PU2024-46	15000	Completed	1251235	13600.38043	NCC	Closed
43	PPSR/2024/Apr/04 Module 2 Tool Stuck up after plug pressing	Ajay Jambhulkar	Production	Anil Kori	PPSR	Yes		10000	Completed	1091672	11866	NCC	Closed
61	PPSR/2024/Jan/09 EAM Lever Welding Incomplete	Suresh Kute	Production	Anil Kori	PPSR	Yes		10000	Completed	990932	10771	NCC	Closed
54	PPSR/2024/May/15 TATA 1.5L Housing Spool valve fitment id unclean during Machining	Ranjit Rupnawar	Quality	Anil Kori	PPSR	Yes		10000	Completed	988632	10746	NCC	Closed
38	PPSR/2024/May/20 FAMB RR1 Test NOK	Chandrasekhar Das	Production	Anil Kori	PPSR	Yes	PU2024-53	10000	Completed	976421	10613.27174	FTQ	Closed
50	PPSR/2024/Nov/05 TATA 3L Housing Steel Ring cut after finished grooving Operation	Atul Raut	Maintenance	Anil Kori	PPSR	Yes		10000	Completed	879428	9559	NCC	Closed
53	PPSR/2024/Jan/19 TATA 1.5L Distance 25 Variation found after Machining	Rajendra Jena	Production	Anil Kori	PPSR	Yes		9000	Completed	877404	9537	NCC	Closed
62	PPSR/2024/Oct/07 BPV Welding Porosity on Spring Carrier	Rupesh Katke	Quality	Anil Kori	PPSR	Yes		7000	Completed	706192	7676	NCC	Closed
46	PPSR/2024/Oct/01 Module 1 Shaft not completely inserted into flap housing	Rupesh Katke	Quality	Anil Kori	PPSR	Yes		5000	Completed	666816	7248	NCC	Closed
56	PPSR/2024/Jan/10 TATA 1.5L Resting Pad Dimension NOK	Rajendra Jena	Production	Anil Kori	PPSR	Yes		6000	Completed	652740	7095	NCC	Closed
64	PPSR/2024/Mar/05 EMEGR 1 All Models Welding Shift	Rajendra Banpure	Production	Anil Kori	PPSR	Yes		5000	Completed	531116	5773	NCC	Closed
59	PPSR/2024/Mar/09 VECV 494 Gasket Port profile NOK @Camera station	Rahul Manjare	Quality	Anil Kori	PPSR	Yes		5000	Completed	510140	5545	NCC	Closed
55	PPSR/2024/Jan/12 TATA 1.5L Sump face unclean after machining	Rajendra Jena	Production	Anil Kori	PPSR	Yes		5000	Completed	483644	5257	NCC	Closed
45	PPSR/2024/Feb/03 EMEGR 1 TATA 5L Valve Disc incomplete pressing	Rupesh Kankekar	Quality	Anil Kori	PPSR	Yes		5000	Completed	482908	5249	NCC	Closed
57	PPSR/2024/Mar/08 TATA 1.5L Housing drilling depth oversize	Ranjit Rupnawar	Quality	Anil Kori	PPSR	Yes		5000	Completed	403236	4383	NCC	Closed
47	PPSR/2024/Jun/11 EMEGR 2 Hyundai Valve Taper Pressing NOK	Vikrant Mahamulkar	Quality	Anil Kori	PPSR	Yes		5000	Completed	382628	4159	NCC	Closed
60	PPSR/2024/Feb/08 AL N Series Hsg Cap Fitment NOK	Ranjit Rupnawar	Quality	Anil Kori	PPSR	Yes		3000	Completed	312616	3398	NCC	Closed
48	PPSR/2024/May/05 E511 VOP Bush Pressing NOK	Chandrasekhar Das	Production	Anil Kori	PPSR	Yes		2000	Completed	259164	2817	NCC	Closed
49	PPSR/2024/Aug/11 Panther ETB Flap not inset into shaft	Rupesh Katke	Quality	Anil Kori	PPSR	Yes		2000	Completed	249780	2715	NCC	Closed

Best PPSR for the month Oct 2024- QA

ORGANIMPROVE
Impactful Change in 100 Days

RHEINMETALL BE Problem Solving Sheet

PPSR No.: PPSR/2024/May/21

Project Title: FTQ Improvement

Project Leader: **Rahul Manjare**

Team Member(s): **Sunil Pradhan**

1

Problem: ISEKI St 100 Leakage nok

Plant: Division power system

Line/Station: EMGR-3 S1000

Product/Component: ISEKI / HCM

Amount Defects: 9

Discovered on: 26.05.2024 C-Shift at 51:00

Discovered by: **Amit Kumar**

Repeat Case: ☐ yes ☒ no

Starts / Photo

2

Central questions

WHAT: exactly is the problem? **Leak value more than 2l/min**

WHERE: exactly is the observed? **Leakage from back side cover plate**

HOW: often exactly has the problem occurred? **09 parts in a shift**

WHEN: exactly was the problem observed? **26-05-24 C-Shift at 51-100**

3

Containment Action

No.	Containment Action	Responsible	Date	Status
1	Informed to machining team leads for containment	guyant	26.5.24	🟢
2				🟡
3				🟡

4a

Cause localization (Ishikawa)

```

graph TD
    A[Leakage Nok  
Max 2l/min] --> B[Stage 0 Pass  
ΔM]
    A --> C[Stage 0 fail  
ΔP]
    B --> D[Stage 1 Pass  
Process]
    B --> E[Stage 1 fail  
Product]
    C --> F[Stage 2 Pass  
Cross Key]
    C --> G[Stage 2 Pass  
Cover]
    C --> H[Stage 2 fail  
Housing Ra-Nok]
    
```

4b

5 x WHY - Adopting the most probable main cause(s) from the Ishikawa

1	2	3
Housing Sealing face Ra Nok		
Ra Spec - 1.6		
Actual - 2.10		
Sealing face Rough		
Tool line marks observed		
Tool Insert Wear out		
Insert worn out before achieving its target life		

5

Target Life - 100, Actual - 70

Measures to eliminate the identified cause(s)

No.	Measure	Responsible	Deadline	Status
1	Informed to machining team leader check tool and part condition after 50 nos produced	Sunil	27.5.24	🟢
2				🟡
3				🟡
4				🟡
5				🟡

6

Effectiveness

Description: Evidence of effectiveness (downtime testing, trials, short-term capability, metrics, etc.)

7

Standardization

Protection of the successful solution (FMEA, Control Plan, instructions, training, etc.)

No.	Standardization	Responsible	Date	Status
1	Part inspection training given to operator and fire crew person	Sunil	27.5.24	🟢
2	Daily 1 part given to lab for Ra value check	Sunil	27.05.24	🟢
3	Tool life alarm qty change 80 to 60 nos	manojkumar	15.06.24	🟢
4				🟡
5				🟡

8

Read Across

Transfer of the solution to other products/processes (Lessons Learned)

No.	Proposal / Activity	Responsible	Deadline
1			
2			

Explanation in case of non-necessity

9

Completed

The problem-solving case must be presented by the project leader to the steering committee and confirmed by the latter. This also serves to relieve the team leader.

Project leader: **Rahul**

Completed on: **27/10/2024**

Steering committee: **Sunil**

Best PPSR for the month Oct 2024- QA

RHEINMETALL BE Problem Solving Sheet		PPSR No.: PPSR/2024/Mar/07	Project Title: FEA Improvement Raktanta Amol Ali	0
1	Definition of the problem Problem: TATA 1.5 CV Bush fallen down while pressing Plant: Division Power System Line/Station: module-1 St-10 Product/Component: Tata 1.5 CV Coilers Amount Defects: 18 Nov 24 Discovered on: 06/02/24 Discovered by: station itself Repeat Case: ☐ yes ☒ no			
	Sketch / Photo: 			
	2 Central questions: The Problem IS (NCR Product/Process) The Problem IS NOT (Comparison with an OK Product/Process) WHAT: exactly is the problem? Bush falling down from tool during pressing. Bush OK during pressing. WHERE: exactly is the observed? St-10 during pressing. HOW: often exactly has the problem occurred? Frequently in a shift. WHEN: exactly was the problem observed? During tool travel to position for pressing.			
	3 Containment: No Containment Action Responsible Date Status 1 100% - Part detected by station fore Sankarsh wancher 06/02/24 2 3			
	4a Cause localization (Ishikawa) Man: unskilled operator, JIS not followed Machine: Fixture alignment, Air pressure low, Fixture clamping, Jerk movement, Vacuum not created, temperature. Material: Bush DO of S, Tool DO of S Methods: Milieu: Measurement: Problem: Bush falling down while pressing.			
4b	5 x WHY - Adopting the most probable main cause(s) from the Ishikawa 1 Vacuum not created. 2 Gap bush guide 3 Part rusting in proper position 4 Air comes out through tool due to tool guide damage.			
	5 Measures to eliminate the identified cause(s) No. Measure Responsible Deadline Status 1 Top tool replaced. Atul 06/02/24 2 3 4 5			
	6 Effectiveness: Description Evidence of effectiveness (dummy testing, trial, short-term capability, metrics, etc.) 10 Before After 5 0 Dec-23 Jan-24 Feb-24 Mar-24 Apr-24 May-24 Jun-24 Jul-24 Aug-24 Sep-24 Oct-24			
	7 Standardization: Protection of the successful solution (FMEA, Control Plan, instructions, training, etc.) Responsible Date Status 1 Check point added on PM check sheet. Atul 15/10/24 2 3 4 5			
	8 Read Across: Transfer of the solution to other products/processes (Lessons Learned) Proposal / Activity Responsible Deadline 1 Email communication done for check point added on PM check sheet. Atul 18/10/24 2 Explanation in case of non-necessity			
9	Complete The problem-solving case must be presented by the project leader to the steering committee and confirmed by the latter. This also serves to relieve the team leader. Raktanta 18/11/2024 Steering committee member			



Measure with date and responsible person



Measure in implementation



Measure implemented



Effectiveness of the measure proven

PPSR



RANJIT RUPANWAR

16



RUPESH KATKE

19



SUDHIR SHINDE

13



Best PPSR of the Monthly



PPSR

Kata Team leader projects



Sr. No.	PPSR NO	Project Leader	MF	Line/ Machine	Model	Station	Cost save per annum	Problem	Corrective Action	CA Status
1	PPSR/2024/Jan/01	Rupesh Katke	MF03	ETB line	Panther	St-50B	60307.35	ETB flow not ok	1) Total flap 13132 nos checked in which 2171 nos flap OD found nok. 2) OD angle of the part found in opposite direction due to loading not proper so implement pokayoke for incorrect loading.	G
2	PPSR/2024/Jan/02	Rupesh Katke	MF01	Line-1	3D15	St-85	60924.66	Emergency test not ok	1) Found burr in emergency valve gallery so hsg segregated in shop 2) Segregation done by supplier side.	G
3	PPSR/2024/Jan/03	Ranjit Rupnawar	MF03	EAM line	All model	St-60	1689108.18	Leakage not ok	1) Found hsg casting crack so firstly segregate in shop and from supplier side. 2) From supplier side found slide cavity welding worn out so filled with welding.	G
6	PPSR/2024/Jan/06	Raju Kundu	MF02	Module-1	Tata 2.2	St-110	397547.32	Water cycle not ok	From supplier side below action taken 1) Due to extra cutting load due to riser extra material double cut added during milling operation.	G
7	PPSR/2024/Jan/07	Rupesh Kanekar	MF02	Act-2	Hyundai Actuator	St-50-60	821277.97	Axle bend after pressing	Circular tool tightening done.	G
9	PPSR/2024/Jan/09	Suresh Kute	MF03	EAM line	All model	St-30	873579.39	Lever welding not ok (Welding incomplete)	Guage implemented with 100% sorting started Dwell fitted by gefit team	G
13	PPSR/2024/Jan/13	Rupesh Katke	MF03	ETB line	Panther	St-10	234098.17	ETB panther bearing pressing not ok	1) Found ID oversize so informed to Incoming QA and total 768 nos hsg checked in which 07 nos hsg found nok (No go pass) 2) From supplier get oversize due to sludge found in spindle collate so cleaned it.	G
14	PPSR/2024/Jan/14	Ratikanta	MF03	ETB line	Panther	St-10	177301.36	Shaft stuck up during pressing	Action from supplier 1) Dia 10 position 0.06 correction done w.r.t (D-E-F) 2) Control chart for dia 10 position 0.06 started (Fixture-1,2,3,4) 3) 100% reration guage implementation started for dia 10 position (Next 3 month)	G
15	PPSR/2024/Jan/15	Dinesh Kuldharan	MF01	Line-2	Panther VP	St-15	816439.89	Leakage not ok for higher side	1) Machining sequence changed 2) 100% inspection started	G
16	PPSR/2024/Jan/16	Sunil Pradhan	M/c	AMS-1	Hyundai valve hsg	NA	1764800.00	CMM dimation and relation guage variation.	1) Ball screw & bearing changed.	G

PPSR Project 2024
Total PPSR closed= 133

Total scrap cost avoidance
INR= 7,55,97,867
EURO= 8,21,716

SFM Agenda (ACT)– Awareness, Coaching and Training

FRIDAY



KAIZEN

5. All implemented Kaizens/ideas are awarded with a gift Coupon in the Monthly All Hand meeting



4. Each recommendation is then either placed in the Rejected or Implemented Box.



3. All Kaizen's hanging on the tree are reviewed weekly on Friday by SFM Team and each idea is assigned for implementation to a SFM member

Name: _____ Emp Code: _____ Line/Department: _____

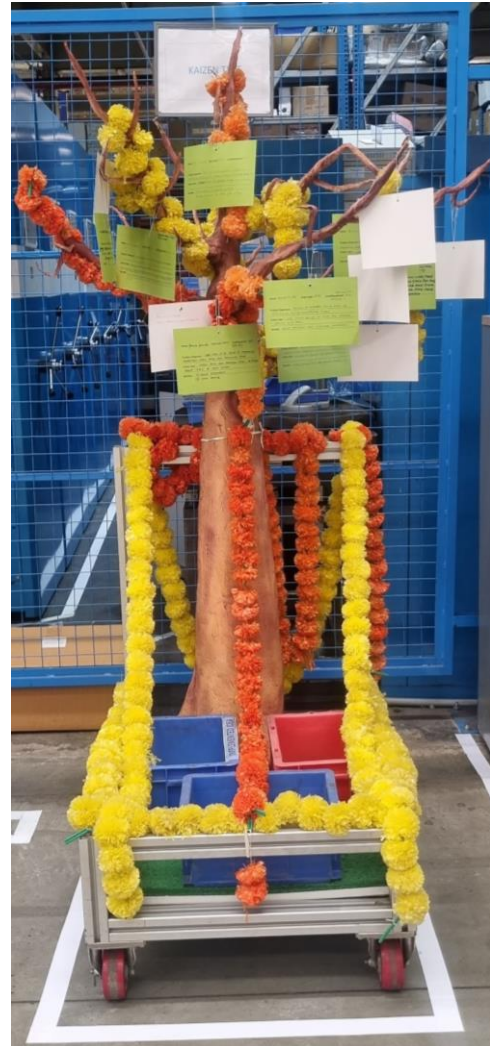
Problem Statement : _____

Action/ Idea : _____

Benefits : _____



2. Fill in the details of the proposed idea/kaizen at hang it on the tree



1. Pick up a blank form from the respective bin

KAIZEN

Kaizen List

Kaizen SR. Number	Month	Suggestion date	Closing target date	Idea Implemented date	Problem/ Before Status	Counter Measure/ After Improvement	Location	Cost save	Benefits in PQCDSM	IDEA By-	Implemented -By	KAIZEN Accept/Reject for Coupon
1	Jan-24	12-Jan-24	27-Jan-24	15-Jan-24	One trolley is empty On cooler station.	Cooler cap is adjusted on this trolley.	Cooler		M	Rajendra Banpure	Rajendra Banpure	YES
2	Jan-24	20-Jan-24	5-Feb-24	21-Jan-24	Near miss happen on st 80 end stopper which is single get loose and actuator tray fallen down.	As shown in figure two stopper added also to restrict height of bin also stopper added so that over capacity tray is avoided.	Emegr-2		Q,S	Rajendra Banpure	Rajendra Banpure	YES
3	Jan-24	19-Jan-24	4-Feb-24	24-Jan-24	Housing lock damage due to the part heating to stopper. (Marked in red box)	Remove the stopper and fix anew stopper to avoid damage.	Act-2	10000	Q,C	Sanjay Shende	Sanjay Shende	
4	Jan-24	15-Jan-24	30-Jan-24	31-Jan-24	Currently there is no provision in HMI to show height not ok . (Mark red in box option not in HMI.)	In HMI this qty showing provision is done so that operator can get easy data while filling production report or during analysis.	Act-2	5000	Q,M,C	Sanjay Shende	Sanjay Shende	YES
5	Jan-24	7-Jan-24	22-Jan-24	12-Jan-24	Cover sheet kept on printer which is unstandardized.	Location is made near station & chances of damage after falling on floor.	Emegr-1	7400	C,M	Rahul Patil	Rahul Patil	YES
6	Jan-24	17-Jan-24	2-Feb-24	20-Jan-24	Module line 1 St 100B scanner charger adapter doesn't have proper location due to this there is chances of adapter damage also it disturb SS standard.	Provided proper location which will doesn't allow to damage adapters pins as well as save damage cost and improves SS.	Module-1	1000	C	Sagar Borade	Sagar Borade	YES
7	Jan-24	13-Jan-24	28-Jan-24	17-Jan-24	Module line-2 st 100 due to part resting pad wear out we have every shift 3-4 adapter scrap for screw thread damage or washout because of screwing gun and adapter thread position mismatch.	Provide one resting pad with dowel with help of industrial engineering after this corrective action there is 0% rejection on module line 2 for screwing nok. FTQ also improves as well as scrap cost zero.	Module-2		Q	Sagar Borade	Sagar Borade	
8	Jan-24	20-Jan-24	5-Feb-24	25-Jan-24	Dowel get loose due continue use and component slightly move while operation that why grade nok & marking not ok rejection is increase.	Dowel dia oversize done and press fit bush is installed and then dowel fixed due to this movement of component hile cycle running is restricted.	Module-1	30000	P,Q,C	Ratha Biswas Sahu	Ratha Biswas Sahu	YES
9	Jan-24	20-Jan-24	5-Feb-24	23-Jan-24	Tool not fit properly in given slot.	Pin is putted in slot so that tool falling and damage is reduced.	Module-1	2000	C,S	Somain sutradhar	Somain sutradhar	
10	Jan-24	2-Jan-24	17-Jan-24	22-Jan-24	As shown door is open and chances of scratch mark while moving in gangway.	Provision of lock is done.	Module-2 backside		S	Sanjay Gupta	Sanjay Gupta	yes
11	Jan-24	10-Jan-24	25-Jan-24	12-Jan-24	Act line-1 st 70 gear cover assembly parts are kept inside the machine due to no proper location.	Made proper location for gear cover assembly parts to keep.	Act-2		M	Rushikesh Suryavanshi	Rushikesh Suryavanshi	YES
12	Jan-24	20-Apr-24	25-Jan-24	24-Apr-24	Act-1 st 80 A after part ok there is no parking station after st 80.	Parking station is made as shown.	Act-1		M	Rushikesh Suryavanshi	Rushikesh Suryavanshi	YES
13	Jan-24	25-Jan-24	10-Feb-24	28-Jan-24	After layout change for parking part for st 90 there is no any location.	As shown parking location is provided as shown in fig.	Module-2		M	Sanket More	Sudipta Bera	YES
14	Jan-24	25-Jan-24	10-Feb-24	30-Jan-24	There is no protection for the QR marking scanner camera and there is a possibility of damage to the camera during part loading and unloading (For Tata plastic cooler PV,CV,2.2)	Protection cover is provided.	Module-1	16000	C	Ratha Biswas Sahu	Ratha Biswas Sahu	YES
15	Jan-24	25-Jan-24	10-Feb-24	28-Jan-24	For st 05 child part kept on Emegr-2 line, every time require to bring from that line.	As shown near station already provided trolley modify so that all child part seat in it.	Act-2		P,M	Abhimanyu Digal	Abhimanyu Digal	YES
16	Feb-24	2-Feb-24	17-Feb-24	12-Feb-24	As shown in figure there is no location for plastic bag.	Location is made at fw table backside.	Emegr-1		M	Rajendra Banpure	Rajendra Banpure	yes
17	Feb-24	9-Feb-24	24-Feb-24	18-Feb-24	Before, There was no interlock was for Gasket present sensor was available so if we do not fix the Gasket the cycle will run. Again we have to rework the part	To saving time we added the animation command so that when bin operate we can check the status of the rejection bin directly on HMI . There is no need to go & check on Manual mode see its position.	Emegr-1 st 50		M	Rajendra Banpure	Rajendra Banpure	
18	Feb-24	16-Feb-24	2-Mar-24	26-Feb-24	Protection cover to sensor not available.	Protection cover provided.	Module-1	8000	C	Dinendra Nath	Dinendra Nath	yes
19	Feb-24	10-Feb-24	24-Feb-24	17-Feb-24	Condition of all rework table is as shown same below.	Soft material sheet is provided and horizontally implemented.	Dismantling St		Q	Nagesh Dandage	Nagesh Dandage	
20	Feb-24	8-Feb-24	23-Feb-24	10-Feb-24	Currently parking station provided bin size of 12 and it will reduce chances to part mix or operation mix component also space occupy.	This bin size reduced to 6 so that online flow will run also chances of mix up part is reducing.	Emegr-2 st 40,30		Q,M	Sanjay Gupta	Sanjay Gupta	

KAIZEN



RATHA SAHU

25



VAIBHAV BARGE

28



SAGAR BORHADE

24

2

1

3

KAIZEN SHEET

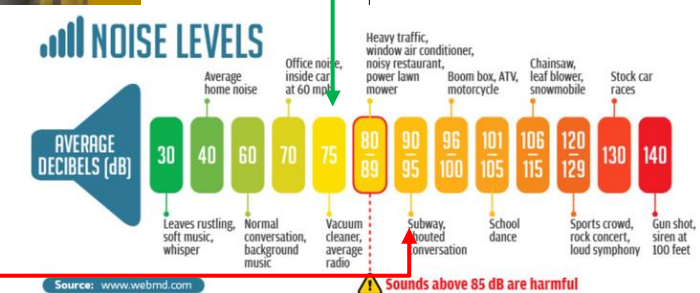
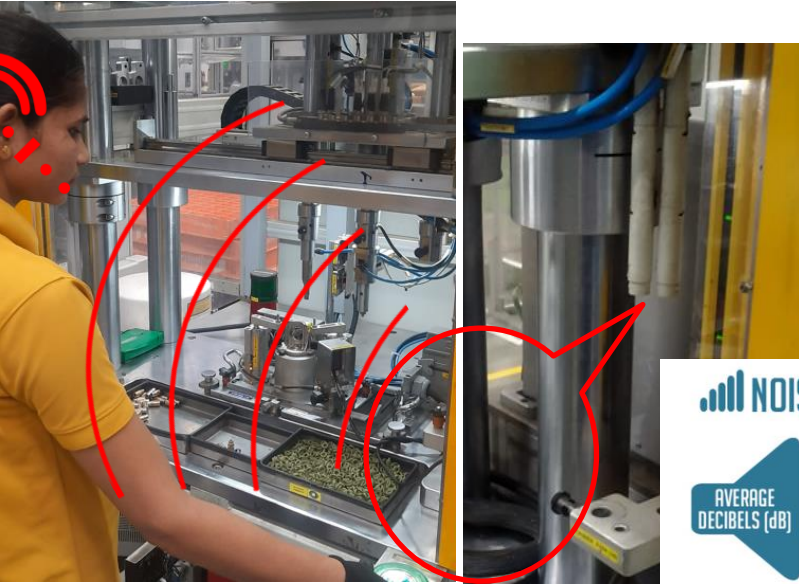
Kaizen Theme: Air flow noise reduction
MF2- EMEGR 2- Station 20

Team: Nilam Dhuri



BEFORE

Excess noise observed on station due to vacuum generator exhaust



Source: www.webmd.com

Sounds above 85 dB are harmful

AFTER

Exhaust pipe has extended to behind station, by which noise has reduced.



KPI	Concern
S	Heavy noise-(109dB) causing 100% use of ear plug.
M	Low- Excess noise during each operating cycle

KPI	Concern
S	Reduced noise level-(75dB) eliminated use of Ear plug.
M	Improved operator working condition.

RESULTS

P

PRODUCTIVITY

↔

Q

QUALITY

↔

C

COST

↔

D

DELIVERY

↔

S

SAFETY

↑

M

MORALE

↑

KAIZEN

Best Kaizen of the Monthly



KAIZEN

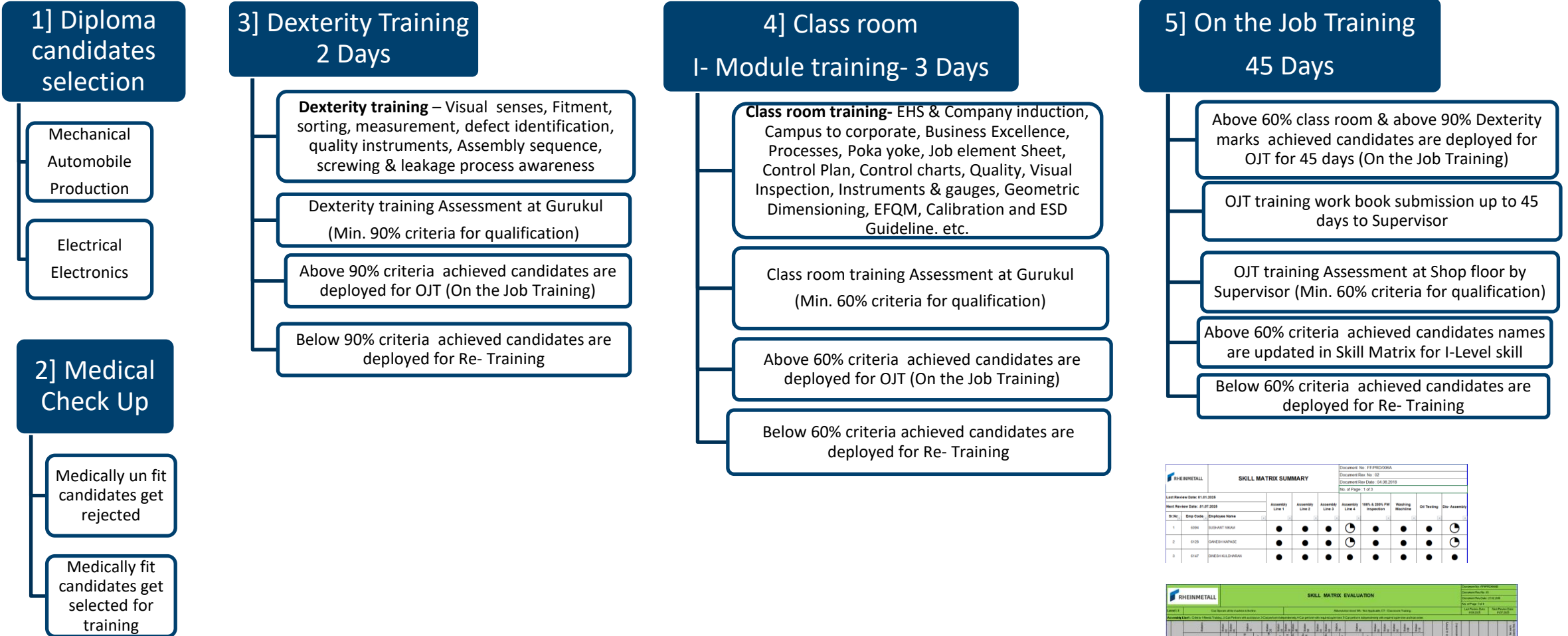
Waste walk and Kaizen - BE Assessment result, Takwe, Pune

Total 441 Kaizen implemented in 2024 and saved INR 11,74,000/-

Value Creation		Topic	0	anchor		2	anchor		4	anchor		Score
				1			3			5		
7	Are waste walks implemented in operations areas focusing on the 8 kinds of waste according to the S+A standards linked to existing Value Streams and done with a team of employees from different levels and functions of the organization? (e.g. operators, line leaders, other functions)?	Waste Walk	Very few random waste walks done. No planning or structure in place.	Waste walks are done independent from value stream on regular basis and are done by leaders only (line leader, shift leader, MF leader, etc.)		Waste walks are done according to the S+A standard linked to the main product family VSMs only. Waste walks are done regularly, improvement actions are defined and tracked. Majority of the waste walks are done with either cross functional or cross hierarchical teams.				Waste walks are fully implemented according to the S+A standard, linked to the current state VSMs for all product families in all areas. Waste walks are done regularly, with teams from either cross functions or cross hierarchies. Improvement actions are defined and tracked. Additional waste walks are conducted randomly with mixed teams depending on situation or purpose. Process is similar defined as the layered audit process. Waste walk process is reviewed once per year and improved.	5	

100% score achieved during 2023 & 2024 BE assessment

New Operating Engineers Training Flow Chart



SKILL MATRIX SUMMARY									
Last Review Date: 01.01.2025									
Next Review Date: 01.07.2025									
Sl. No.	Shop Code	Employee Name	Assessment Line 1	Assessment Line 2	Assessment Line 3	Assessment Line 4	100% & 200% PM Inspection	Working Machine	On Testing
1	0004	RESHANT KUMAR	●	●	●	●	●	●	●
2	0108	SAVESH KAPASE	●	●	●	●	●	●	●
3	0147	SURESH KULSHARNAN	●	●	●	●	●	●	●

SKILL MATRIX EVALUATION									
Last Review Date: 01.01.2025									
Next Review Date: 01.07.2025									
Sl. No.	Shop Code	Employee Name	Assessment Line 1	Assessment Line 2	Assessment Line 3	Assessment Line 4	100% & 200% PM Inspection	Working Machine	On Testing
1	0004	RESHANT KUMAR	●	●	●	●	●	●	●
2	0108	SAVESH KAPASE	●	●	●	●	●	●	●
3	0147	SURESH KULSHARNAN	●	●	●	●	●	●	●

Skill Upgrade Training (L, U and O) Flow Chart

6] L- Module Training 3 days

L- Module training- 5's, Kaizen, Hydraulics and Pneumatics, 8D_Fishbone_Why Why, CNC, Tools and Preset, TPM, PLC, Troubleshooting, IATF, Behavioral etc awareness

Candidates are worked on shop floor approx. one years or depends on his/her performance, deployed for L- Module training

L- level skill is evaluated by shop floor supervisor and qualified candidates skill matrix is updated

During evaluation, not qualified candidates are deployed for Re-Training

SKILL MATRIX SUMMARY									
Document No: FF-PRD0006A									
Last Review Date: 01.01.2025									
Next Review Date: 01.01.2026									
Sl No.	Emp Code	Employee Name	Assembly Line 1	Assembly Line 2	Assembly Line 3	Assembly Line 4	100% & 200% PM Inspection	Welding Machine	Oil Testing
1	6004	SHARAD KUMAR	●	●	●	●	●	●	●
2	6108	SHARAD KUMAR	●	●	●	●	●	●	●
3	6147	SHARAD KUMAR	●	●	●	●	●	●	●

SKILL MATRIX EVALUATION									
Document No: FF-PRD0006A									
Last Review Date: 01.01.2025									
Next Review Date: 01.01.2026									
Sl No.	Emp Code	Employee Name	Assembly Line 1	Assembly Line 2	Assembly Line 3	Assembly Line 4	100% & 200% PM Inspection	Welding Machine	Oil Testing
1	6004	SHARAD KUMAR	●	●	●	●	●	●	●
2	6108	SHARAD KUMAR	●	●	●	●	●	●	●
3	6147	SHARAD KUMAR	●	●	●	●	●	●	●

7] U- Module Training Class room- 3 days

U- Module training- MSA,SPC, APQP, PPAP, PFMEA, SFM, SMED, Problem solving, TPM, Process orientation, Customer orientation, Analytical Thinking & Problem solving, Establishing Focus, Result orientation etc awareness

Candidates are worked on shop floor approx. two years or depends on his/her performance, deployed for U- Module training

U- level skill is evaluated by shop floor supervisor and qualified candidates skill matrix is updated

During evaluation, not qualified candidates are deployed for Re-Training

SKILL MATRIX SUMMARY									
Document No: FF-PRD0006A									
Last Review Date: 01.01.2025									
Next Review Date: 01.01.2026									
Sl No.	Emp Code	Employee Name	Assembly Line 1	Assembly Line 2	Assembly Line 3	Assembly Line 4	100% & 200% PM Inspection	Welding Machine	Oil Testing
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2	6108	SHARAD KUMAR	●	●	●	●	●	●	●
3	6147	SHARAD KUMAR	●	●	●	●	●	●	●

SKILL MATRIX EVALUATION									
Document No: FF-PRD0006A									
Last Review Date: 01.01.2025									
Next Review Date: 01.01.2026									
Sl No.	Emp Code	Employee Name	Assembly Line 1	Assembly Line 2	Assembly Line 3	Assembly Line 4	100% & 200% PM Inspection	Welding Machine	Oil Testing
1	6004	SHARAD KUMAR	●	●	●	●	●	●	●
2	6108	SHARAD KUMAR	●	●	●	●	●	●	●
3	6147	SHARAD KUMAR	●	●	●	●	●	●	●

8] O- Module Training Class room- 3 days

O- Module training- Job shadowing and Supervisory development training awareness

Candidates are worked on shop floor approx. three years or depends on his/her performance, deployed for O- Module training

O- level skill is evaluated by shop floor supervisor and qualified candidates skill matrix is updated

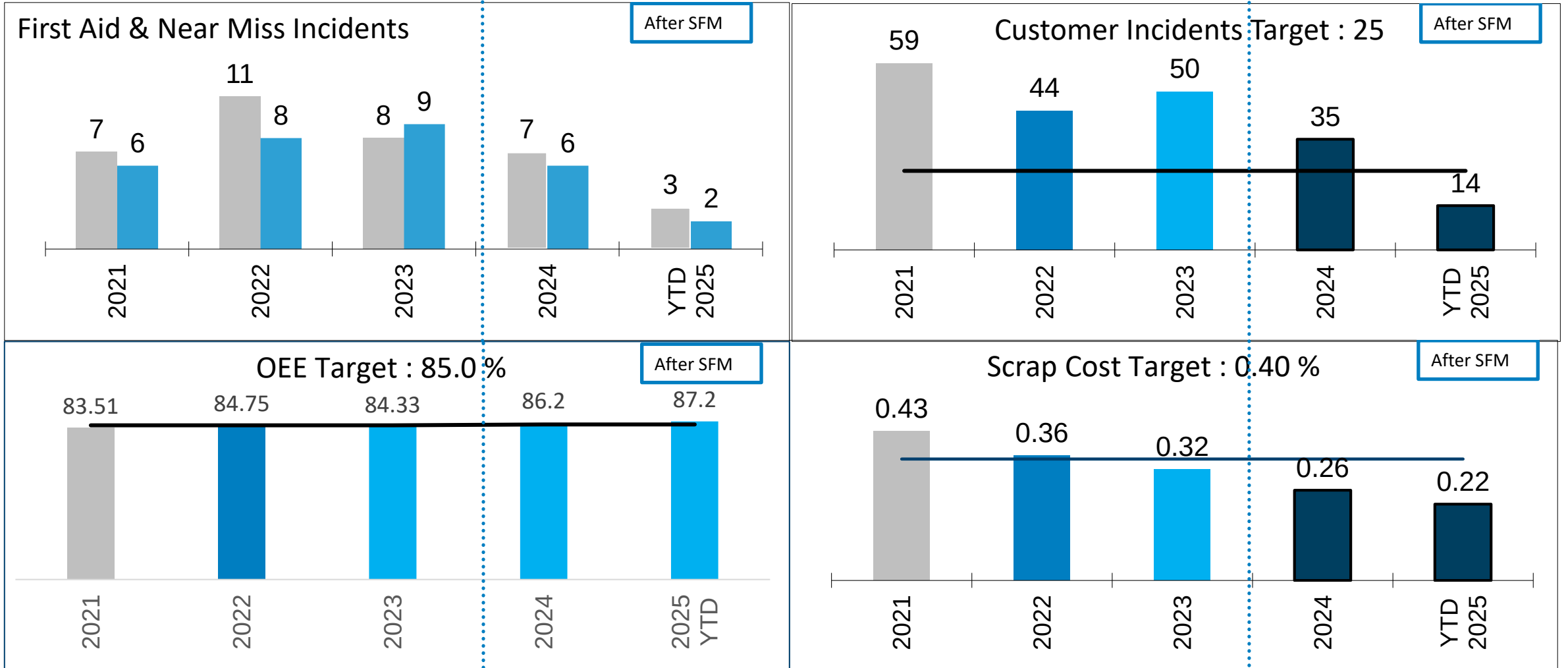
During evaluation, not qualified candidates are deployed for Re-Training

SKILL MATRIX SUMMARY									
Document No: FF-PRD0006A									
Last Review Date: 01.01.2025									
Next Review Date: 01.01.2026									
Sl No.	Emp Code	Employee Name	Assembly Line 1	Assembly Line 2	Assembly Line 3	Assembly Line 4	100% & 200% PM Inspection	Welding Machine	Oil Testing
1	6004	SHARAD KUMAR	●	●	●	●	●	●	●
2	6108	SHARAD KUMAR	●	●	●	●	●	●	●
3	6147	SHARAD KUMAR	●	●	●	●	●	●	●

SKILL MATRIX EVALUATION									
Document No: FF-PRD0006A									
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1	6004	SHARAD KUMAR	●	●	●	●	●	●	●
2	6108	SHARAD KUMAR	●	●	●	●	●	●	●
3	6147	SHARAD KUMAR	●	●	●	●	●	●	●

Benefits & Impact

- Quantitative: Fewer breakdowns, better OEE, higher productivity
- Qualitative: Ownership, Culture shift, Faster problem-solving



Top 50 best performer celebration



TAKING RESPONSIBILITY IN A CHANGING WORLD